

Mensuration – I

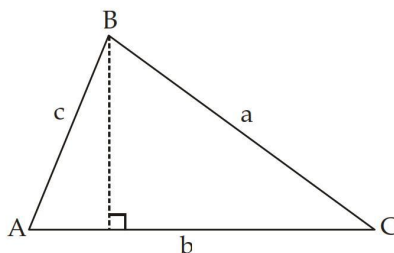
Mensuration is the branch of Mathematics that deals with the measurement of length, area and volume.

In this chapter, we will study the areas and perimeters of two-dimensional figures.

Area: The area of any plane figure is the amount of surface enclosed within its boundary lines. It is expressed in square units e.g. square meter, square centimeter, square inch, etc.

Perimeter: The perimeter of a plane figure is the total length of the sides enclosing the figure.

Area of Δ : The area of a triangle is represented by the symbol Δ . For a triangle ABC, the three sides are represented by a, b and c and the angles opposite to these sides are represented by A, B and C respectively, as shown in the figure :



Let us look at the formulae for areas and perimeters of various plane figures.

(i) For any triangle in general,

(a) When base (b) and height (h) perpendicular to that base are given,

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} b \cdot h$$

(b) When lengths of all three sides are given a, b and c.

$$\text{Area} = \sqrt{S(S-a)(S-b)(S-c)} \text{ where, } S = \frac{a+b+c}{2} = \text{semi-perimeter of the triangle.}$$

This is called **Hero's Formula**.

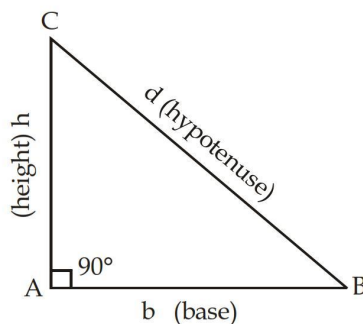
(c) If the lengths of three medians of a triangle ABC are p, q, r units, then:

$$\text{Area} = \frac{4}{3} \sqrt{S_m(S_m-p)(S_m-q)(S_m-r)}$$

$$\text{where, } S_m = \frac{p+q+r}{2}$$

(d) Perimeter = $a + b + c = 2S$

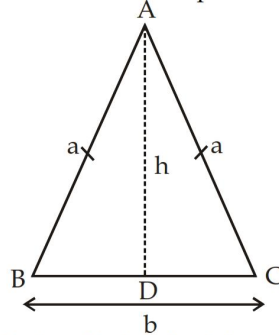
(ii) **Right Angled triangle:**



(a) $\text{Area} = \frac{1}{2} \times \text{Product of the sides containing the right angle}$

(b) $d^2 = b^2 + h^2$

(iii) **Isosceles triangle:** A triangle in which two sides are equal is said to be an isosceles triangle.



(a) Area = $\frac{b}{4}\sqrt{4a^2 - b^2}$ where 'a' is the length of each of the two equal sides and b is the third side.

(b) Perimeter = $2s = 2a + b$

(c) Height (h) = $\frac{\sqrt{4a^2 - b^2}}{2}$

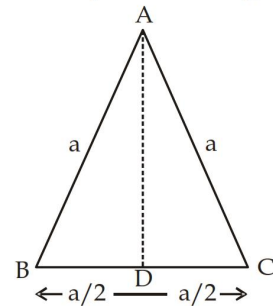
(iv) **Equilateral triangle:** A triangle in which all three sides are equal is said to be an equilateral triangle.

(a) Area = $\frac{\sqrt{3}}{4}a^2$, where 'a' is the side of the triangle.

(b) Height of an equilateral triangle (h) = $\frac{\sqrt{3}}{2}a$

(c) Perimeter (P) = $3a$

(d) $\angle A = \angle B = \angle C = 60^\circ$



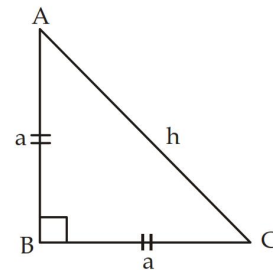
(v) **Isosceles right-angled triangle**

(i) $AB = BC = a$ and $\angle B = 90^\circ$

(ii) $h = a\sqrt{2}$

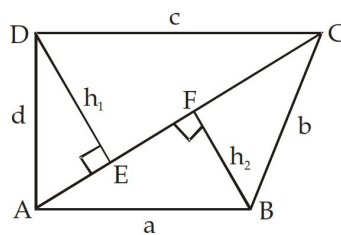
(iii) Area = $\frac{1}{2}a^2$

(iv) Perimeter = $2a + h = 2S$



QUADRILATERALS

A closed figure bounded by four sides is called as quadrilateral.



(i) For any quadrilateral in general,

(a) $\angle A + \angle B + \angle C + \angle D = 360^\circ$

(b) Area of the Quadrilateral = $\frac{1}{2} \times (\text{one diagonal}) \times (\text{sum of perpendiculars to it from opposite vertex})$

$$= \frac{1}{2} \times (AC) \times (h_1 + h_2)$$

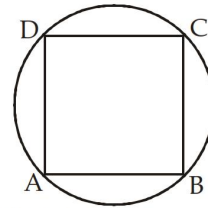
(c) Perimeter = $a + b + c + d$.

- (ii) For a cyclic Quadrilateral where the four sides measure a, b, c and d respectively.

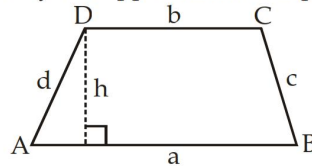
$$\text{Area} = \sqrt{(S - a)(S - b)(S - c)(S - d)}$$

$$\text{where } S \text{ is the semi-perimeter, i.e., } S = \frac{a + b + c + d}{2}$$

$$\angle A + \angle C = \angle B + \angle D = 180^\circ$$



- (iii) **Trapezium:** A quadrilateral where any two opposite sides are parallel is a Trapezium

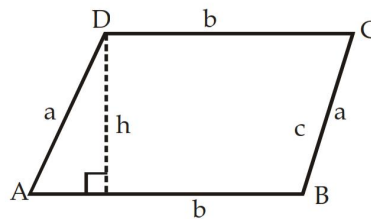


(a) Area of a trapezium = $\frac{1}{2} \times (\text{sum of parallel sides}) \times (\text{perpendicular distance between them})$

$$= \frac{1}{2}(a + b) \times h$$

(b) Perimeter (P) = $a + b + c + d$.

- (iv) **Parallelogram:** A parallelogram is a quadrilateral whose opposite sides are equal and parallel.



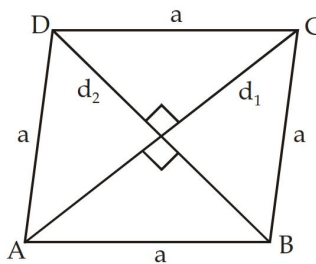
(a) Area = base $\times$ height.

(b) Area = Product of two sides $\times$ sine of the angle between two adjacent sides = $ab \sin \theta$.

(c) Perimeter = $2(a + b)$

(d) $d_1^2 + d_2^2 = 2(a^2 + b^2)$ (d_1, d_2 = length of diagonals)

- (v) **Rhombus:** It is a parallelogram whose all four sides are equal

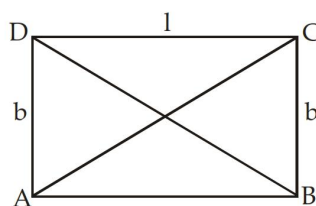


(a) Diagonals of a Rhombus bisect each other at 90°

(b) Area = $\frac{1}{2} \times$ product of the diagonals.

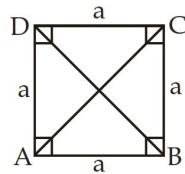
(c) perimeter = $4 \times$ side of the rhombus.

- (vi) **Rectangle**



- (a) Area = Length $\times$ Breadth.
- (b) Perimeter = $2(l + b)$, where l and b are the length and the breadth of the rectangle respectively.
- (c) Diagonals are equal and bisect each other.

(vii) **square**

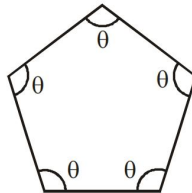


All four sides of a square are equal and all four vertex angles are equal to 90° .

- (a) Area = (side)²
- (b) Area = $\frac{1}{2} \times (\text{Diagonal})^2$ [where diagonal = $\sqrt{2} \times \text{side}$]
- (c) Perimeter = $4 \times \text{side}$.

POLYGONS: A polygon is a plane figure enclosed by four or more straight lines.

Regular Polygon: A polygon whose all sides are equal is called a Regular polygon.



All the interior angles of a regular polygon are equal.

For a regular polygon:

- (i) Sum of exterior angles = 360°
- (ii) Sum of interior angles = $(n - 2) \times 180^\circ$
- (iii) Interior angle + exterior angle = 180°
- (iv) Each interior angle = $\frac{(n - 2)}{n} \times 180^\circ$
- (v) Each exterior angle = $\frac{360^\circ}{n}$
- (vi) No. of diagonals in a polygon = $\frac{n(n - 3)}{2}$
- (vii) Area of a regular polygon = $\frac{1}{2} \times (\text{Perimeter}) \times (\text{Perpendicular distance from the centre of the polygon to any side})$

(Centre of a regular polygon is equidistant from all its sides).

Area of regular Hexagon = $6 \times (\text{Area of equilateral triangle of side } a)$

Since regular Hexagon is made of 6 equilateral triangle when its vertex are joined to centre.

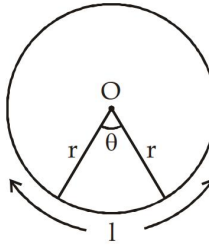
$$= 6 \times \frac{\sqrt{3}}{4} a^2 = \frac{3\sqrt{3}}{2} a^2$$

Regular octagon: Area = $2(\sqrt{2} + 1)(\text{side})^2$

CIRCLE

- (i) Area of circle = πr^2 , where r is the radius of the circle
- (ii) Circumference of a circle = $2\pi r$

Sector of a circle:



$$\text{Length of arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2$$

where θ is the angle of the sector in degrees and r is the radius of the circle.

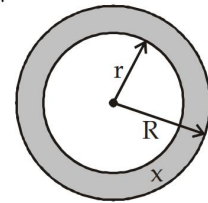
Ring or circular path: Ring is the space enclosed between two concentric circles.

$$\text{Area} = \pi(R^2 - r^2)$$

$$\text{Perimeter} = 2\pi(R + r)$$

Pathways running across the middle of a rectangle.

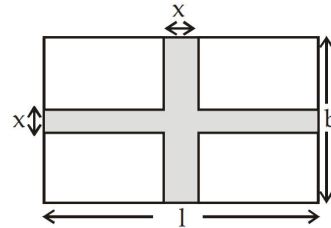
x = width of the path.



Pathway Across the rectangle

$$\text{Area of path} = (l + b - x)x$$

$$\text{Perimeter of path} = 2(l + b) - 4x = 2(l + b - 2x)$$

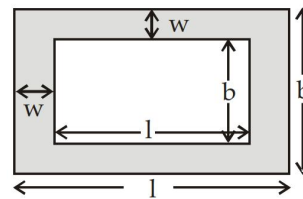


Pathways around a rectangular space:

Outer pathways:

$$(i) \text{ Area} = (l + b + 2w)2w$$

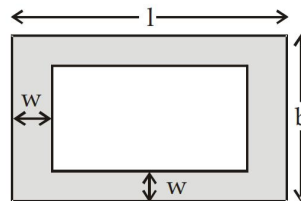
$$(ii) \text{ Perimeter} = \text{outer perimeter} + \text{inner perimeter} \\ = 2(l + b) + 2(l + b + 4w) = 4(l + b + 2w).$$



Inner Pathways:

$$\text{Area of the path} = (l + b - 2w)2w$$

$$\text{Perimeter} = 2(l + b) + 2(l + b - 4w) = 4(l + b - 2w)$$



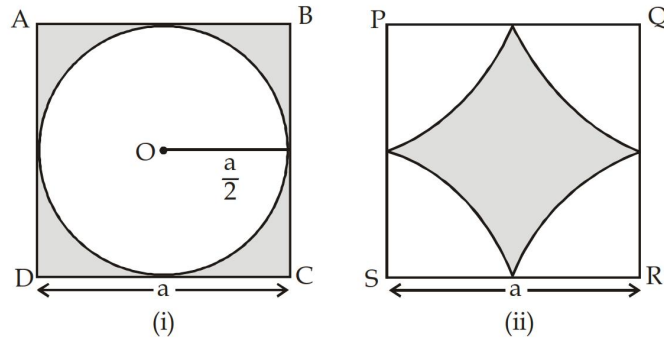
Room: If we have a room of length l , breadth b and height h , then,

Area of four walls of the room = $2h(l + b)$

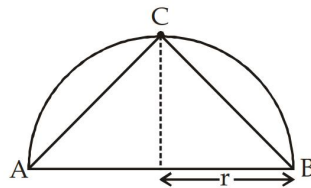
Area of four walls and floor = $2h(l + b) + lb$

Area of floor, roof and four walls = $2h(l + b) + 2lb$.

Some important results:



1. If the diagonal of a square becomes x times, then the area of the square becomes x^2 times.
2. If each of the defining dimensions or sides of any 2-D figure are increased (or decreased) by $x\%$, its perimeter also increases (or decreases) by $x\%$.
3. If all the sides of a quadrilateral are increased (or decreased) by $x\%$, its diagonals also increase (or decrease) by $x\%$.
4. If the area of a square is $a \text{ cm}^2$, then the area of the circle having circumference equal to the perimeter of the square is $= \frac{4a}{\pi} \text{ cm}^2$.
5. In figure (i) ABCD is a square of side 'a' and O is the centre of the inscribed circle with radius $a/2$ in figure (ii) PQRS is a square of side 'a' and P, Q, R and S are the centres of four quadrants of circle with radius $\frac{a}{2}$. Area of shaded region in both cases $= \frac{3}{14} a^2$.
6. The area of largest triangle inscribed in a semi-circle of radius r is r^2 .



Area of $\triangle ACB = r^2$

7. If each of the defining dimensions or sides of any 2-D figure is changed by $x\%$, its area changes by

$$x \left(2 + \frac{x}{100} \right) \%$$

if the dimension is increased, take x as positive.

if the dimension is decreased, take x as negative.

8. If the length of a rectangle is increased by $a\%$, then its breadth will have to be decreased by $\left(\frac{100a}{100 + a} \right) \%$ in order to maintain the same area of the rectangle.

9. If the length and breadth of a rectangle are increased by $x\%$ and $y\%$ respectively, then the area of rectangle will increase by: $\left(x + y + \frac{xy}{100}\right)\%$
10. If the length and breadth of a rectangle are decreased by $x\%$ and $y\%$ respectively, then the area of the rectangle will decrease by: $\left(x + y - \frac{xy}{100}\right)\%$.
11. The number of revolutions made by a circular wheel of radius r in travelling a distance 'd' is given by:

$$n = \frac{d}{2\pi r}$$

Types of Questions

1. The area of a triangle whose sides are 15m, 16m and 17m is:

Sol. $S = \frac{15 + 16 + 17}{2} = 24$

$$\text{Area} = \sqrt{S(S-a)(S-b)(S-c)} \\ = \sqrt{24(24-15)(24-16)(24-17)} = 24\sqrt{21} \text{ m}^2.$$

2. Find the area of an isosceles right-angled triangle whose hypotenuse is 16 cm :

Sol. hypotenuse = $\sqrt{2}a = 16$

$$a = \frac{16}{\sqrt{2}} = 8\sqrt{2} \text{ cm.}$$

$$\text{Area} = \frac{1}{2}a^2 = \frac{1}{2} \times 8\sqrt{2} \times 8\sqrt{2} = 64 \text{ cm}^2.$$

3. A rectangular plot is $116 \times 68 \text{ m}^2$. It has a gravel path of 2.5 m wide all around in it. Find the area of the gravel path?

Sol. Area of the gravel path = $(l + b - 2w) 2w$
 $= (116 + 68 - 5)5 = 895 \text{ m}^2$

4. Find the area of a rhombus one of whose diagonals measures 8cm and the other 10 cm?

Sol. Area = $\frac{1}{2}d_1d_2 = \frac{1}{2} \times 8 \times 10 = 40 \text{ cm}^2$.

5. The length of a rectangle is increased by 60%. By what percent would the width be decreased so as to maintain the same area?

Sol. If the length of a rectangle is increased by $a\%$, then its breadth will have to be decreased by $\left(\frac{100a}{100+a}\right)\%$ in order to maintain the same area of the rectangle.

$$\text{So, required value} = \frac{100 \times 60}{100 + 60} = 37\frac{1}{2}\%$$

6. The area of the largest triangle that can be inscribed in a semi circle of radius x in square unit is:

Sol. The area of the largest triangle that can be inscribed in a semi-circle of radius x is x^2 unit.

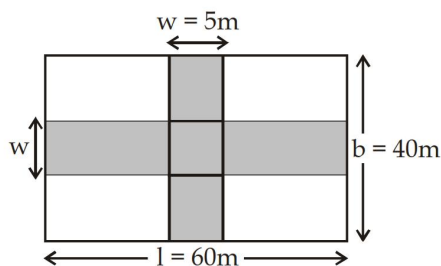
7. A circle and a rectangle have the same perimeter. The sides of the rectangle are 18 cm and 26cm. The area of the circle is:

Sol. $2\pi r = 2(18 + 26)$

$$2 \times \frac{22}{7} \times r = 44 \times 2 \Rightarrow r = 14 \text{ cm.}$$

$$\text{Area of the circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

8. A rectangular lawn $60 \times 40 \text{ m}^2$ has two roads each 5m wide running between the park. One is parallel to length and other is parallel to width. Cost of gravelling is 60 paise/ m^2 . Find the total cost of gravelling of the roads.



Sol. Area of path = $(l + b - w)w = (60 + 40 - 5)5 = 475 \text{ m}^2$.

$$\text{Total cost} = 475 \times \frac{60}{100} = \text{Rs } 285$$

9. Area of circle is equal to the area of a rectangle having perimeter of 100 cms and length is more than the breadth by 6cms. What is the diameter of the circle?

Sol. Let breadth = b

$$\text{then, length} = b + 6$$

$$\Rightarrow 2(b + 6 + b) = 100 \Rightarrow b = 22 \text{ cm} \Rightarrow l = 28 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = 28 \times 22$$

$$\Rightarrow \frac{22}{7} \times r^2 = 28 \times 22 \Rightarrow r = 14 \text{ cm.}$$

$$\text{Diameter} = 28 \text{ cm.}$$

10. A metal wire when bent in the form of a square encloses an area 484 cm^2 . If the same wire is bent in the form of a circle, then (taking $\pi = \frac{22}{7}$) its area is

Sol. Let the side of the square be a

$$\Rightarrow a^2 = 484 \text{ cm}^2. \Rightarrow a = 22 \text{ cm.}$$

$$\Rightarrow \text{Perimeter of square} = 4a = 88 \text{ cm} = \text{circumference of circle}$$

$$\Rightarrow 2\pi r = 88 \text{ cm.} \Rightarrow 2 \times \frac{22}{7} \times r = 88 \text{ cm.}$$

$$\Rightarrow r = 14 \text{ cm.}$$

$$\text{So, area of circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2.$$

Foundation

Questions

1. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is:
 (a) 96 (b) 128
 (c) 143 (d) 144
2. If the perimeter of a rectangular field is 200 m and its breadth is 40 m then its area is (in m^2):
 (a) 1200 (b) 2400
 (c) 4800 (d) 6000
3. Around a rectangular garden of length 10 m and width 5 m, a road 1 m wide is laid. The cost of metalling the road at the of Rs. 200 per m^2 is:
 (a) Rs. 3400 (b) Rs. 6000
 (c) Rs. 6800 (d) Rs. 13200
4. The cost of levelling a rectangular ground at Rs. 1.25 per sq. metre is Rs. 900. If the length of the ground is 30 metres, then the width is:
 (a) 330 metres (b) 34 metres
 (c) 24 metres (d) 18 metres
5. The length of a rectangle is twice its breadth. If its length is decreased by 5 cm and breadth is increase by 5 cm, the area of the rectangle is increased by 75 cm^2 . Then the length of the rectangle is:
 (a) 20 cm (b) 30 cm
 (c) 40 cm (d) 50 cm
6. A 5 m wide lawn is cultivated all along the outside of a rectangular plot measuring $90 \text{ m} \times 40 \text{ m}$. The total area of the lawn is:
 (a) 1200 m^2 (b) 1300 m^2
 (c) 1350 m^2 (d) 1400 m^2
7. If the diagonal of a rectangle is 17 cm long and the perimeter of the rectangle is 46cm, the area of the rectangle is:
 (a) 120 cm^2 (b) 112 cm^2
 (c) 132 cm^2 (d) 289 cm^2
8. The dimensions of the floor of a rectangular hall are $4\text{m} \times 3\text{m}$. The tiles floor of the hall is to be tiled with regular tiles with $8 \text{ cm} \times 6 \text{ cm}$ dimension without breaking tiles to smaller size. The number of tiles required:
 (a) 4800 (b) 2600
 (c) 2500 (d) 2400
9. The number of marble slabs of size $20 \text{ cm} \times 30 \text{ cm}$ required to pave the floor of a square room of side 3m, is:
 (a) 150 (b) 100
 (c) 25 (d) 225
10. If the side of square is doubled, then the ratio of area of resulting square to that of the given square is:
 (a) 1 : 2 (b) 2 : 1
 (c) 3 : 1 (d) 4 : 1
11. If the length of diagonal of a square is 5.2 cm, the area of square is:
 (a) 15.12 cm^2 (b) 13.52 cm^2
 (c) 12.62 cm^2 (d) 10 cm^2
12. The number of square tin sheets of side 20 cm that can be cut off from a square tin of side 1m, is:
 (a) 5 (b) 25
 (c) 50 (d) 100
13. A coloured cloth 1.75 metre long and 105 cm wide is made in to a square handkerchiefs of side 35 cm. The number of handkerchief of side 35 cm.
 (a) 10 (b) 15
 (c) 25 (d) 35
14. If the side of a square is increased by 50%. The percent increase in area is:
 (a) 50 (b) 100
 (c) 125 (d) 150
15. The ratio of the area of a square to that of the square drawn on its diagonal is:
 (a) 1 : 1 (b) 1 : 2
 (c) 1 : 3 (d) 1 : 4

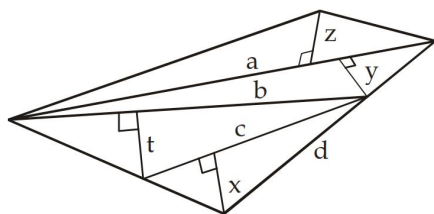
16. A triangle of area $9 \times y \text{ cm}^2$ has been drawn such that its area is equal to the area of an equilateral triangle of side 6 cm. Then the value of y is:

(a) $\sqrt{2} \text{ cm}$ (b) $\sqrt{3} \text{ cm}$
(c) 2 cm (d) 3 cm

17. If the altitude of an equilateral triangle is $\sqrt{6}$, then its area is:

(a) $3\sqrt{3}$ (b) $2\sqrt{3}$
(c) $2\sqrt{2}$ (d) $6\sqrt{2}$

18. A surveyor in his field book has drawn the plot as shown in the given figure. The area of the plot is:



(a) $\frac{1}{2}(az + by + ct + dx)$
(b) $\frac{1}{2}(bt + cx + ay + az)$
(c) $\frac{1}{2}(cx + bt + by + az)$
(d) $\frac{1}{2}(d + t)(c + x) + \frac{1}{2}(a + b)(y + z)$

19. If the circumference of a circle is 352 metres, then its area in m^2 is:

(a) 9856 (b) 8956
(c) 6589 (d) 5986

20. If the circumference of a circle is 88 cm, then the area of a circle (in sq. cm) is:

(a) 196π (b) 92π
(c) 64π (d) 48π

21. The area of a regular hexagon whose one side is 2 m, is:

(a) $3\sqrt{3} \text{ m}^2$ (b) $2\sqrt{3} \text{ m}^2$
(c) $6\sqrt{3} \text{ m}^2$ (d) $4\sqrt{3} \text{ m}^2$

22. Circular quadrants of unit radius are cut from the corners of a rectangle of length 10 m and width 2π metres. The area of the remaining portion is:

(a) $\left(\frac{39}{2}\pi\right) \text{ m}^2$ (b) $\left(\frac{77}{4}\pi\right) \text{ m}^2$
(c) $\left(\frac{35}{2}\pi\right) \text{ m}^2$ (d) $(19\pi) \text{ m}^2$

23. The ratio of the areas of the incircle and circumcircle of square is:

(a) $1:\sqrt{2}$ (b) $1:\sqrt{3}$
(c) $1:4$ (d) $1:2$

24. If the area of a circle be equal to that of a square, then the ratio of the side of the square and the radius of the circle is:

(a) $\sqrt{\pi}:1$ (b) $1:\sqrt{\pi}$
(c) $1:\pi$ (d) $\pi:1$

25. If a piece of wire 20 cm long is bent into an arc of a circle subtending an angle of 60° at the centre, then the radius of the circle (in cms) is:

(a) $\frac{\pi}{120}$ (b) $\frac{\pi}{60}$
(c) $\frac{120}{\pi}$ (d) $\frac{60}{\pi}$

26. The ratio of the length of a side of an equilateral triangle and its height is:

(a) $2:1$ (b) $1:2$
(c) $2:\sqrt{3}$ (d) $\sqrt{3}:2$

27. If the perimeter of a rhombus is $4a$ and length of its diagonals are x and y , then its area is:

(a) $a(x + y)$ (b) $x^2 + y^2$
(c) xy (d) $\frac{1}{2}xy$

28. The sides AB and BC of a parallelogram are 35 m and 50 m respectively. If the length of perpendicular drawn from A to BC is 20 m, then the area of parallelogram is:

(a) 1000 m^2 (b) 1750 m^2
(c) 700 m^2 (d) 400 m^2

29. The area of a triangle whose sides are 15 m, 16 m and 17 m is:

(a) $24\sqrt{4} \text{ m}^2$ (b) $24\sqrt{3} \text{ m}^2$
(c) $24\sqrt{21} \text{ m}^2$ (d) None of these

30. Three sides of a triangular field are 20m, 21 m and 29m long, respectively. The area of the field is:

(a) 215 m^2 (b) 230 m^2
(c) 210 m^2 (d) None of these

31. The hypotenuse and the semi-perimeter of right triangle are 20 cm and 24 cm, respectively. The other two sides of the triangle are:

(a) 16 cm, 12 cm (b) 14 cm, 16 cm
(c) 20 cm, 16 cm (d) None of these

32. Find the area of an isosceles right-angled triangle whose hypotenuse is 8 cm.
 (a) 32 cm^2 (b) 24 cm^2
 (c) 16 cm^2 (d) None of these
33. The area of a parallelogram is 240 cm^2 and its height is 12 cm. The base of the parallelogram is:
 (a) 24 cm (b) 20 cm
 (c) 28 cm (d) None of these
34. If the ratio of the areas of two squares is 9 : 1, the ratio of their perimeters is:
 (a) 9 : 1 (b) 3 : 4
 (c) 3 : 1 (d) 1 : 3
35. The cost of carpeting a room 5 m wide with carpet at Rs. 3.50 per m^2 is Rs. 105. The length of the room is:
 (a) 3.5 m (b) 5 m
 (c) 6 m (d) 6.5 m
36. The diameters of two concentric circles are 8 cm and 10 cm. The area of the region between them is:
 (a) $\pi \text{ cm}^2$ (b) $3\pi \text{ cm}^2$
 (c) $6\pi \text{ cm}^2$ (d) $9\pi \text{ cm}^2$
37. There are two squares s_1 and s_2 . The ratio of their areas is 4 : 25. If the side of s_1 is 6 cm, what is the side of s_2 ?
 (a) 20 cm (b) 15 cm
 (c) 5 cm (d) 12 cm
38. The length of a rectangular hall is $\frac{4}{3}$ of its width. If the area of the hall is 300 m^2 , what is the difference between the length and breadth?
 (a) 15 m (b) 20 m
 (c) 3 m (d) 5 m
39. The diameter of a circle is 105 cm less than the circumference. What is the diameter of circle?
 (a) 44 cm (b) 46 cm
 (c) 48 cm (d) 49 cm
40. Area of the cyclic square is 256 cm^2 , then what is the area of the circle inscribing the square?
 (a) $64\pi \text{ cm}^2$ (b) $118\pi \text{ cm}^2$
 (c) $136\pi \text{ cm}^2$ (d) $128\pi \text{ cm}^2$

Moderate

1. A rectangular plot of land with an area 600 m^2 is fenced, length of fencing being 100 m. Length of the plot is:
 (a) 120 m (b) 25 m
 (c) 30 m (d) 35 m
2. The length and breadth of a rectangular field are 120 m and 80 m respectively. Inside the field, a park of 12m width is made around the field. The area of the park is:
 (a) 3224 m^2 (b) 7344 m^2
 (c) 2256 m^2 (d) 4224 m^2
3. If the perimeter of a rectangle and a square each is equal to 80 cm, and the difference of their areas is 100 sq. cm, the sides of the rectangle are:
 (a) 35 cm, 15 cm (b) 30 cm, 10 cm
 (c) 25 cm, 15 cm (d) 28 cm, 12 cm
4. The area of a right angled triangle is 30 cm^2 and the length of its hypotenuse is 13 cm. The length of the shorter leg is:
 (a) 4 cm (b) 5 cm
 (c) 6 cm (d) 7 cm
5. A plot of land is in the shape of a right angle isosceles triangle. The length of the hypotenuse is $50\sqrt{2} \text{ m}$. The cost of fencing it at rate of Rs. 3 metre will be:
 (a) less than Rs. 300 (b) less than Rs. 400
 (c) more than Rs. 500 (d) more than Rs. 600
6. The perimeter of an isosceles triangle is equal to 14 cm, the lateral side is to the base in the ratio 5 : 4. The area of the triangle is:
 (a) $\frac{1}{2}\sqrt{21} \text{ cm}^2$ (b) $\frac{3}{2}\sqrt{21} \text{ cm}^2$
 (c) $\sqrt{21} \text{ cm}^2$ (d) $2\sqrt{21} \text{ cm}^2$
7. A square and an equilateral triangle have equal perimeters. If the diagonal of the square is $12\sqrt{2} \text{ cm}$, then the area of the triangle is:
 (a) $24\sqrt{2} \text{ cm}^2$ (b) $24\sqrt{3} \text{ cm}^2$
 (c) $48\sqrt{3} \text{ cm}^2$ (d) $64\sqrt{3} \text{ cm}^2$
8. ABCD is a quadrilateral, $AC = 19 \text{ cm}$. The length of perpendicular from B and D on AC are 5 cm and 7 cm respectively. Then, the area of ABCD in sq. cm is:
 (a) 228 (b) 114
 (c) 104 (d) 57
9. A wire is in the form of an equilateral triangle with area 5 m^2 . If it is changed into a circle, the radius will be:
 (a) $\frac{3\sqrt{20}}{2\pi} \text{ m}$ (b) $3^{\frac{3}{4}} \frac{\sqrt{20}}{\pi} \text{ m}$
 (c) $\sqrt{3} \frac{\sqrt{20}}{2\pi} \text{ m}$ (d) $3^{\frac{3}{4}} \frac{\sqrt{20}}{2\pi} \text{ m}$

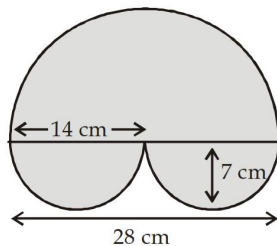
10. Let the areas of the inscribing and the circumscribing circle of equilateral triangle of side $2a$ be A and B . Then?

- (a) $A = \frac{1}{3}\pi a^2$, $B = \frac{4}{3}\pi a^2$
 (b) $A = \frac{1}{4}\pi a^2$, $B = \frac{4}{3}\pi a^2$
 (c) $A = \frac{1}{3}\pi a^2$, $B = \pi a^2$
 (d) $A = \frac{1}{2}\pi a^2$, $B = \frac{4}{3}\pi a^2$

11. If a square $ABCD$ is inscribed in a circle and $AB = 4$ cms, the radius of the circle is:

- (a) 2 cms (b) $2\sqrt{2}$ cms
 (c) 4 cms (d) $4\sqrt{2}$ cms

12. In the given figure, the area of the shaded region (in cm^2), is:



- (a) 324 (b) 428
 (c) 462 (d) 500

13. A circle of radius a is divided into 6 equal sectors. An equilateral triangle is drawn on the chord of each sector to lie outside the circle. Area of the resulting figure is:

- (a) $\frac{3\sqrt{3}}{2}\pi a^2$ (b) $\frac{3\sqrt{3}}{2}a^2$
 (c) $3\pi a^2$ (d) $3\sqrt{3}a^2$

14. The length of a side of rhombus is 5 m and length of one of its diagonals is 8 m. The other diagonal is of length:

- (a) 5 m (b) 6 m
 (c) 7 m (d) 8 m

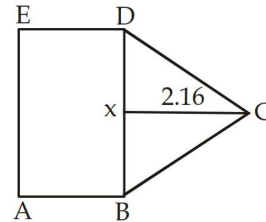
15. A parallelogram has sides 60 m and 40 m and one of its diagonal is 80 m long. Then, its area is:

- (a) 480 m^2 (b) 320 m^2
 (c) $600\sqrt{15}\text{ m}^2$ (d) $450\sqrt{15}\text{ m}^2$

16. $ABCD$ is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. If its diagonals intersect each other at O , then ratio of areas of triangles AOB and COD is:

- (a) 1 : 2 (b) 2 : 1
 (c) 1 : 4 (d) 4 : 1

17. In the given figure, $AB = BD = DE = EA = 2.5$ cm. If $BX = XD$, then the area of the figure (in sq. cm) is:



- (a) 8.95 cm^2 (b) 15.40 cm^2
 (c) 12.70 cm^2 (d) 9.50 cm^2

18. The sides of a triangle are in the ratio 3 : 4 : 5. If its perimeter is 36 cm, area of the triangle is:

- (a) 57 cm^2 (b) 54 cm^2
 (c) 56.5 cm^2 (d) None of these

19. If the area of a triangle with base x is equal to the area of a square with side x , then the altitude of the triangle is:

- (a) $\frac{x}{2}$ (b) x
 (c) $2x$ (d) $3x$

20. If the area of a triangle is 150 m^2 and base: height is 3:4, find its height:

- (a) 25 m (b) 35 m
 (c) 20 m (d) None of these

21. The adjacent sides of a parallelogram are 8 m and 5m. The distance between the longer sides is 4 m. The distance between the shorter sides is:

- (a) 4.6 m (b) 6.4 m
 (c) 8.6 m (d) None of these

22. The area of a parallelogram is 72 cm^2 and its altitude is twice the corresponding base. The length of the base is:

- (a) 6 cm (b) 8 cm
 (c) 8 cm (d) None of these

23. Find the perimeter of a circular plot which occupies an area of 154 cm^2 .

- (a) 54 m (b) 44 m
 (c) 22 m (d) 11 m

24. What is the radius of a circle, to the nearest cm, whose area is equal to the sum of the areas of three circles of radii 22 cm, 19 cm and 8 cm, respectively?

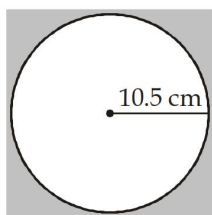
- (a) 17 cm (b) 30 cm
 (c) 29 cm (d) 19 cm

25. A rectangular carpet has an area of 120 m^2 and a perimeter of 46 m. The length of its diagonal is:

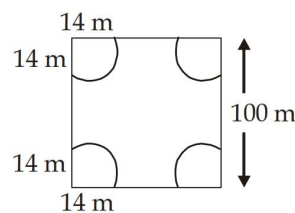
- (a) 15 m (b) 16 m
(c) 17 m (d) 20 m
26. A rectangular carpet has an area of 60 m^2 . Its diagonal and longer side together equal 5 times the shorter side. The length of the carpet is:
(a) 5 m (b) 13 m
(c) 14.5 m (d) 12 m
27. If the diagonal of a square is doubled to make another square, the area of the new square will:
(a) become four-fold (b) become two-fold
(c) become six-fold (d) become eight-fold
28. A garden is 24 m long and 14 m wide. There is a path 1 m wide outside the garden along its sides. If the path is to be constructed with square marble tiles $20 \text{ cm} \times 20 \text{ cm}$, find the number of tiles required to cover the path?
(a) 1800 (b) 200
(c) 2000 (d) 2150
29. The perimeter of an isosceles trapezium is 16 cm and the shorter parallel side is one-third of the longer parallel side. If the sum of the lengths of the opposite sides of the trapezium are equal, find the area (in cm^2) of the trapezium?
(a) $8\sqrt{15}$ (b) $4\sqrt{15}$
(c) $12\sqrt{15}$ (d) $16\sqrt{15}$
30. A iron rod is in the form of a square enclosing an area of 22 cm^2 . If the same iron rod is bent into a circle, then find the area of that circle?
(a) 24 cm^2 (b) 26 cm^2
(c) 28 cm^2 (d) 30 cm^2
31. A rectangular plot of grass is 80 metres long and 60m broad. From 3 metre wide path goes across the centre from the opposite side. Find the area of these path?
(a) 420 m^2 (b) 411 m^2
(c) 402 m^2 (d) None of these
32. The radius of a circular wheel is 1.75 m. The number of revolutions that it will make in covering 11 kms is:
(a) 1000 (b) 100000
(c) 100 (d) 10
33. Diagonal of a rectangle is 25 cm and length is 20 cm. Find the area of rectangle?
- (a) 200 cm^2 (b) 400 cm^2
(c) 500 cm^2 (d) 300 cm^2
34. The length and breadth of a rectangular plot are increased by 10% and 5%. By what percent will its area increase?
(a) 15.5% (b) 16.5%
(c) 17.5% (d) 14.5%
35. A garden is 75 m long and 60 m wide. There is a way 5 m wide parallel to length of garden in its mid. Find the area of the way?
(a) 300 m^2 (b) 325 m^2
(c) 375 m^2 (d) 400 m^2
36. The length of a rectangle is increased by 60%. By what percent should the width be decreased to maintain the same area?
(a) $33\frac{1}{2}\%$ (b) $35\frac{1}{2}\%$
(c) $37\frac{1}{2}\%$ (d) $39\frac{1}{2}\%$
37. The area of a circle is halved when its radius is decreased by n. Find its radius?
(a) $\frac{\sqrt{2n}}{\sqrt{2}-1}$ (b) $\frac{\sqrt{3n}}{\sqrt{4}-1}$
(c) $\frac{\sqrt{4n}}{\sqrt{2}-1}$ (d) $\frac{\sqrt{2n}}{\sqrt{4}-1}$
38. A circular wire, of radius 56 cm is bent into the shape of a square. Find the side of the square?
(a) 44 cm (b) 22 cm
(c) 88 cm (d) 176 cm
39. There are two paths each 10 m wide and cut each other perpendicularly at the mid of a garden 150 m long and 120 m wide. Find the cost of gravelling the path at Rs. 2/ m^2 .
(a) Rs. 5200 (b) Rs. 4800
(c) Rs. 5600 (d) Rs. 4400
40. The cost of fencing a square garden is Rs. 400 at the rate of Rs. 1 per metre. The area of the garden in square metres is:
(a) 10 (b) 10^2
(c) 10^3 (d) 10^4

Difficult

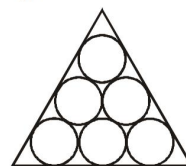
1. A man builds a circular pool of radius 5 m inside a circular garden of radius 12 m. In order to compensate the area lost due to the construction of pool, he extends the radius 'r' m while keeping the garden still circular, so that the area of the garden remains the same. The value of r (in m) is:
(a) 1 (b) $\sqrt{5}$
(c) $\sqrt{12} - 5$ (d) $\frac{5}{\pi}$
2. What is the area of the shaded portion if each side of the square measures 21 cm?



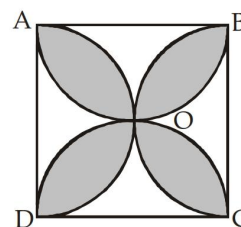
- (a) 86.5 cm^2 (b) 102 cm^2
(c) 94.5 cm^2 (d) 81.5 cm^2
3. A circle is inscribed in an equilateral triangle of side 8 cm. The area of the portion between the triangle and the circle is:
- (a) 11 cm^2 (b) 10.95 cm^2
(c) 10 cm^2 (d) 10.50 cm^2
4. Mid-points of a triangle containing sides 12 cm, 18 cm and 26 cm are joined to each other. Find the area of triangle formed by the mid-points?
- (a) $4\sqrt{5} \text{ cm}^2$ (b) $2\sqrt{7} \text{ cm}^2$
(c) $4\sqrt{35} \text{ cm}^2$ (d) $8\sqrt{35} \text{ cm}^2$
5. A piece of ground measures 19 meters 2.5 dm by 12 metres 5 dm. From the centre of each side a path 2 meters wide goes across to the centre of the opposite side. Find the cost of paying these paths at the rate of Rs . 12.32 per sq. meter?
- (a) Rs. 7430.68 (b) Rs. 713.04
(c) Rs. 753.26 (d) Rs. 733.04
6. In the given figure, $OP = PQ = 7 \text{ cm}$ and semicircles are drawn on OP , PQ and OQ as diameters. Then, the perimeter of the shaded area is:
-
- (a) 88 cm (b) 176 cm
(c) 264 cm (d) 352 cm
7. The circumference of a circular field is 440 m. A circular path of 10 metre width runs around the outside of the field. Find the cost of cultivating the path at the rate of 70 paise per square metre?
- (a) Rs. 2200 (b) Rs. 3300
(c) Rs. 264 (d) None of these
8. A square park has each side 100 m. At each corner of the park there is a flower bed in the form of a quadrant of radius 14 m, as shown in the figure. The area of the remaining part of the park is:



- (a) 9384 m^2 (b) 9290 m^2
(c) 9150 m^2 (d) 9050 m^2
9. Perimeter of Rhombus is $2P$ unit and sum of length of diagonals is m unit then the area of the Rhombus is?
- (a) $\frac{1}{4}m^2P$ sq. unit (b) $\frac{1}{4}mP^2$ sq. unit
(c) $\frac{1}{4}(m^2 - P^2)$ sq. unit (d) $\frac{1}{4}(m^2 + P^2)$ sq. unit
10. A rectangular park $60 \times 40 \text{ m}^2$ has two cross roads running in the middle of the park and the rest park has been lawn. If the area of the lawn is 2109 m^2 . What is the width of the road?
- (a) 3 m (b) 4 m
(c) 5 m (d) 7 m
11. An equilateral triangle circumscribes all the circles, each with radius 10 cm. What is the perimeter of the equilateral triangle?



- (a) $20(2 + \sqrt{3}) \text{ cm}$ (b) $30(2 + \sqrt{3}) \text{ cm}$
(c) $60(2 + \sqrt{3}) \text{ cm}$ (d) None of these
12. In the given figure ABCD is a square. Four equal semicircles are drawn in such a way that they meet each other at 'O'. Sides AB, BC, CD and DA are the respective diameters of the four semicircles. Each of the side of the square is 8 cm. Find the area of the shaded region?

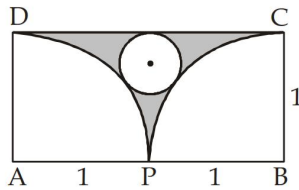


- (a) $32(\pi - 2) \text{ cm}^2$ (b) $16(\pi - 2) \text{ cm}^2$
(c) $(2\pi - 8) \text{ cm}^2$ (d) $\left(\frac{3}{4}\pi - 4\right) \text{ cm}^2$

13. Find the perimeter of a square which is symmetrically inscribed in a semicircle of radius 10 cm?

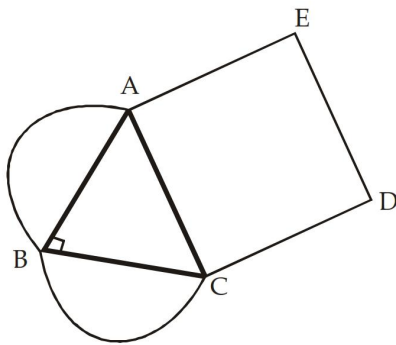
(a) $\sqrt{80}$ cm (b) 80 cm
(c) $8\sqrt{24}$ cm (d) $16\sqrt{5}$ cm

14. In the following figure ABCD is a rectangle with AD and DC equal to 1 and 2 units respectively. Two quarter circles are drawn with centres at B and A respectively. Now a circle is drawn touching both the quarter circles and the side of the rectangle. Find the area of the shaded region:



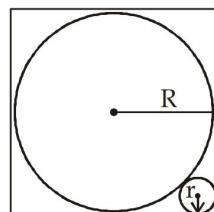
(a) $\frac{32}{115}$ square units (b) $\frac{13}{56}$ square units
(c) $\frac{16}{83}$ square units (d) $\frac{7}{20}$ square units

15. The area of the square on AC as a side is 128 cm^2 . What is the sum of the areas of semicircles drawn on AB and AC as diameters, given ABC is an isosceles right angled triangle and AC is its hypotenuse?



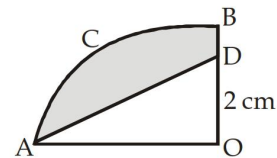
(a) $32\pi \text{ cm}^2$ (b) $16\pi \text{ cm}^2$
(c) 16 cm^2 (d) 32 cm^2

16. In the given figure, find the radius of smaller circle (r):



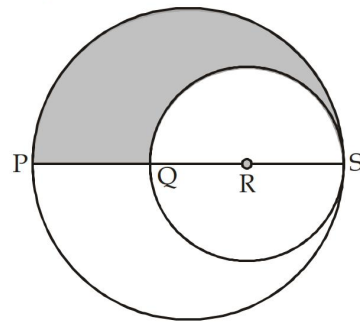
(a) $(\sqrt{3} - 2\sqrt{2})R$ (b) $2(\sqrt{2} - 1)R$
(c) $(3 - 2\sqrt{2})R$ (d) None of these

17. In the adjoining figure, AOB is a quadrant of a circle of radius 3.5 cm with centre O. Calculate the area of the shaded portion? (Use $\pi = 22/7$)



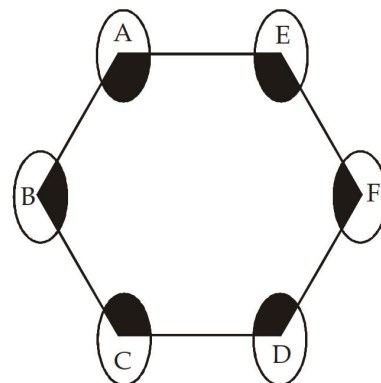
(a) 35 cm^2 (b) 7.875 cm^2
(c) 9.625 cm^2 (d) 6.125 cm^2

18. PQRS is the diameter of a circle of radius 6 cm. The lengths PQ, QR and RS are equal. Semi-circles are drawn with PQ and QS as diameters as shown in the figure alongside. Find the ratio of the area of the shaded region to that of the unshaded region?



(a) 1 : 2 (b) 25 : 121
(c) 5 : 18 (d) 5 : 13

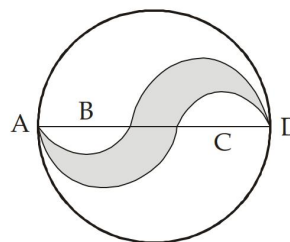
19. Find the sum of the areas of the shaded sectors given that ABCDEF is any hexagon and all the circles are of same radius r with different vertices of the hexagon as their centres as shown in the figure?



(a) πr^2 (b) $2\pi r^2$
(c) $\frac{5\pi r^2}{4}$ (d) $\frac{3\pi r^2}{2}$

20. In the given figure, B and C are points on the diameter AD of the circle such that $AB = BC = CD$. Then find the ratio of area of the shaded portion to that of the whole circle?

- (a) 1 : 3 (b) 2 : 3
(c) 1 : 2 (d) None of these



Previous Year (Memory Based)

- A square lawn has a path 4 m wide around it. If the area of the path is 196 m^2 , then each side of the lawn is:
(a) 7.25 m (b) 9.25 m
(c) 8.25 m (d) 8 m
- The diagonal of a square is $4\sqrt{2} \text{ cm}$. The diagonal of another square, whose area is double that of the first square, is:
(a) $8\sqrt{2} \text{ cm}$ (b) 16 cm
(c) $\sqrt{32} \text{ cm}$ (d) 8 cm
- If the diagonals of two squares are in the ratio of 2 : 5, their areas will be in the ratio of:
(a) $\sqrt{2} : \sqrt{5}$ (b) 2 : 5
(c) 4 : 25 (d) 4 : 5
- Each side of a rectangular field is diminished by 40%. By how much percent is the area of the field diminished?
(a) 32 (b) 64
(c) 25 (d) 16
- A path of uniform width runs round the inside of a rectangular field 38 m long and 32 m wide. If the path occupies 600 m^2 , then the width of the path is:
(a) 30 m (b) 5 m
(c) 18.75 m (d) 10 m
- The length of a plot is five times its breadth. A playground measuring 245 m^2 occupies half of the total area of the plot. What is the length of the plot?
(a) $35\sqrt{2} \text{ m}$ (b) $175\sqrt{2} \text{ m}$
(c) 490 m (d) $5\sqrt{2} \text{ m}$
- If the lengths of the sides of a triangle are in the ratio 4 : 5 : 6 and the inradius of the triangle is 3 cm, then the altitude of the triangle corresponding to the largest side as base is:
(a) 10 cm (b) 8 cm
(c) 7.5 cm (d) 6 cm
- The lengths of two sides of an isosceles triangle are 15 and 22, respectively. What are the possible values of perimeter?
(a) 52 or 59 (b) 52 or 60
(c) 15 or 37 (d) 37 or 29
- If the altitude of a triangle is increased by 10% while its area remains same, its corresponding base will have to be decreased by:
(a) 10% (b) 9%
(c) $9\frac{1}{11}\%$ (d) $11\frac{1}{9}\%$
- If D and E are the mid-points of the sides AB and AC, respectively of the $\triangle ABC$ what per cent of the whole triangular region is the area of trapezium BCED?
(a) 50% (b) 25%
(c) 75% (d) 60%
- If the area of a triangle is 1176 cm^2 and base : corresponding altitude is 3 : 4, then the altitude of the triangle is:
(a) 42 cm (b) 52 cm
(c) 54 cm (d) 56 cm
- The area of a field in the shape of a trapezium measures 1440 m^2 . The perpendicular distance between its parallel sides is 24 cm. If the ratio of the parallel sides is 5 : 3, the length of the longer parallel side is:
(a) 75 m (b) 45 m
(c) 120 m (d) 60 m
- The area of a circle is increased by 22 cm^2 when its radius is increased by 1 cm. The original radius of the circle is:
(a) 3 cm (b) 5 cm
(c) 7 cm (d) 9 cm
- A semi-circular shaped window has diameter of 63 cm. Its perimeter equals: $\left(\text{take } \pi = \frac{22}{7} \right)$
(a) 126 cm (b) 162 cm
(c) 198 cm (d) 251 cm
- A 7 m wide road runs outside around a circular park, whose circumference is 176 m. The area of the road is: $\left(\text{use } \pi = \frac{22}{7} \right)$

- (a) 1386 m^2 (b) 1472 m^2
(c) 1512 m^2 (d) 1760 m^2
16. If the difference between the circumference and diameter of a circle is 30 cm, then the radius of the circle must be:
(a) 6 cm (b) 7 cm
(c) 5 cm (d) 8 cm
17. The ratio of the outer and the inner perimeters of a circular path is 23 : 22. If the path is 5 m wide, the diameter of the inner circle is:
(a) 110 m (b) 55 m
(c) 220 m (d) 230 m
18. Three coins of the same size (radius 1 cm) are placed on a table such that each of them touches the other two. The area enclosed by the coins is:
(a) $\left(\frac{\pi}{2} - \sqrt{3}\right) \text{ cm}^2$ (b) $\left(\sqrt{3} - \frac{\pi}{2}\right) \text{ cm}^2$
(c) $\left(2\sqrt{3} - \frac{\pi}{2}\right) \text{ cm}^2$ (d) $\left(3\sqrt{3} - \frac{\pi}{2}\right) \text{ cm}^2$
19. Four equal circles each of radius 'a' units touch one another. The area enclosed between them in square units, is $\left(\text{take } \pi = \frac{22}{7}\right)$
(a) $3a^2$ (b) $\frac{6a^2}{7}$
(c) $\frac{41a^2}{7}$ (d) $\frac{a^2}{7}$
20. The area of the ring between two concentric circles, whose circumferences are 88 cm and 132 cm, is:
(a) 780 cm^2 (b) 770 cm^2
(c) 715 cm^2 (d) 660 cm^2
21. In an equilateral triangle of side 24 cm, a circle is inscribed touching its sides. The area of the remaining portion of the triangle is $\left(\sqrt{3} = 1.732\right)$
(a) 98.55 sq cm (b) 100 sq cm
(c) 101 sq cm (d) 95 sq cm
22. A horse is tied to a pole fixed at one corner of a $30 \text{ m} \times 30 \text{ m}$ square field of grass, by means of a 10 m long rope. Find the area of the field, in which the horse can graze? $\left(\text{take } \pi = 3.14\right)$
(a) 78.5 sq m (b) 79.5 sq m
(c) 76.0 sq m (d) 78.0 sq m
23. Area of an equilateral triangle is $4\sqrt{3} \text{ sq cm}$. Then the length of the diagonal of a square whose side is equal to the height of the equilateral triangle, (in cm), is:
(a) 4 (b) $4\sqrt{6}$
(c) $2\sqrt{6}$ (d) $2\sqrt{3}$
24. An equilateral triangle is drawn on the diagonal of a square. The ratio of the area of the triangle to that of the square is:
(a) $\sqrt{3} : 2$ (b) $\sqrt{2} : \sqrt{3}$
(c) $2 : \sqrt{3}$ (d) $1 : \sqrt{2}$
25. A circle and a square have equal areas. The ratio of a side of the square and the radius of the circle is:
(a) $1 : \sqrt{\pi}$ (b) $\sqrt{\pi} : 1$
(c) $1 : \pi$ (d) $\pi : 1$
26. The perimeter of a rectangle and a square are 160 m each. The area of the rectangle is less than that of the square by 100 sq m. The length of the rectangle is:
(a) 30 m (b) 60 m
(c) 40 m (d) 50 m
27. The areas of a square and a rectangle are equal. The length of the rectangle is greater than the length of a side of the square by 5 cm and the breadth is less than the length of the side of the square by 3 cm. The perimeter of the rectangle is:
(a) 17 cm (b) 26 cm
(c) 30 cm (d) 34 cm
28. There is a rectangular tank of length 180 m and breadth 120 m in a circular field. If the area of the land portion of the field is 40000 m^2 , what is the radius of the field? $\left(\text{take } \pi = \frac{22}{7}\right)$
(a) 130 m (b) 135 m
(c) 140 m (d) 145 m
29. The perimeter of a rectangular table is 28m., whereas its area is 48 m^2 . What is the length of its diagonal?
(a) 5 m (b) 10 m
(c) 12 m (d) 12.5 m
30. A wire when bent in the form of a square encloses an area of 484 sq. cm . What will be the enclosed area when the same wire is bent in the form of a circle?
 $\left(\text{take } \pi = \frac{22}{7}\right)$
(a) 125 cm^2 (b) 230 cm^2
(c) 550 cm^2 (d) 616 cm^2
31. The perimeter of five squares are 24 cm, 32 cm, 40 cm, 76 cm and 80 cm respectively. The perimeter of another square equal in area to sum of the areas of these squares is:
(a) 31 cm (b) 62 cm
(c) 124 cm (d) 961 cm

32. A path of uniform width runs round the inside of a rectangular field 38 m long and 32 m wide. If the path occupies 600 m^2 , then the width of the path is:
 (a) 30 m (b) 5 m
 (c) 18.75 m (d) 10 m
33. The area of a rectangle is thrice that of a square. The length of the rectangle is 20 cm and the breadth of the rectangle is $\frac{3}{2}$ times that of the side of the square.
 The side of the square, (in cm) is:
 (a) 10 (b) 20
 (c) 30 (d) 60
34. The length of a room is 3 m more than its breadth. If the area of a floor of the room is 70 m^2 , then the perimeter of the floor will be:
 (a) 14 m (b) 28 m
 (c) 34 m (d) 17 m
35. A can go round a circular path 8 times in 40 minutes. If the diameter of the circle is increased to 10 times the original diameter, the time required by A to go round the new path once travelling at the same speed as before is:
 (a) 25 min (b) 20 min
 (c) 50 min (d) 100 min
36. The perimeter of a triangle is 30 cm and its area is 30 cm^2 . If the largest side measures 13 cm, What is the length of the smallest side of the triangle?
 (a) 3 cm (b) 4 cm
 (c) 5 cm (d) 6 cm
37. Find the diameter of a wheel that makes 113 revolutions to go 2 km 26 decameters? (Take $\pi = \frac{22}{7}$)
 (a) $4\frac{4}{13} \text{ m}$ (b) $6\frac{4}{11} \text{ m}$
 (c) $12\frac{4}{11} \text{ m}$ (d) $12\frac{8}{11} \text{ m}$
38. If the circumference of a circle increases from 4π to 8π , what change occurs in its area?
 (a) It doubles (b) It triples
 (c) It quadruples (d) It is halved
39. If the circumference and area of a circle are numerically equal, then the diameter is equal to:
 (a) area of the circle (b) $\pi/2$
 (c) 2π (d) 4
40. The length of the floor of room exceeds its breadth by 20m. The area of the floor remains unaltered when the length is decreased by 10 m but the breadth is increased by 50 m. The area of the floor (in square meters) is:
 (a) 280 (b) 325
 (c) 875 (d) 420

Foundation

Solutions

1. (c); Let the breadth be x
 then length = $x + 2$
 Perimeter of rectangle = $2(l + b)$
 $48 = 2(x + 2 + x)$, $24 = 2x + 2$, $2x = 22$, $x = 11$
 breadth, $x = 11$
 length, $x + 2 = 13$
 Area = $l \times b = 11 \times 13 = 143 \text{ cm}^2$
2. (b); Breadth = 40m, length = x
 Perimeter = $2(x + 40)$
 $200 = 2(x + 40)$, $x + 40 = 100$, $x = 60$
 Area = $40 \times 60 = 2400 \text{ m}^2$
3. (c); Area of road = $2w(l + b + 2w)$
 $= 2(1)(15 + 2) = 34 \text{ m}^2$
 Cost of metalling the road = $200 \times 34 = \text{Rs. } 6800$
4. (c); Area of ground = $\left(\frac{900}{1.25}\right) \text{ m}^2 = 720 \text{ m}^2$
5. (c); Let the breadth be ' x ', length be ' $2x$ '
 Area = $2x^2$
 New length = $(2x - 5) \text{ cm}$
 New breadth = $(x + 5) \text{ cm}$
 New Area = $(2x - 5)(x + 5)$
 ATQ, $2x^2 + 75 = (2x - 5)(x + 5) \Rightarrow x = 20$
 length = $2 \times x = 2 \times 20 = 40 \text{ cm}$
6. (d); Area of lawn = $2w(l + b + 2w)$
 $= 2 \times 5 \times (90 + 40 + 10) = 1400 \text{ m}^2$
7. (a); Let the length and breadth be l and b respectively.
 According to question
 $l^2 + b^2 = 17^2$, $l^2 + b^2 = 289$... (i)
 and $2(l + b) = 46$, $l + b = 23$... (ii)
 From (i) and (ii)
 $(l + b)^2 = l^2 + b^2 + 2lb$
 $lb = \frac{1}{2}(23^2 - 289)$, $lb = 120 \text{ cm}^2$

8. (c); Number of tiles = $\frac{400 \times 300}{8 \times 6} = 2500$

9. (a); Number of marble slabs = $\frac{300 \times 300}{20 \times 30} = 150$

10. (d); Let the original side be x and the new side be $2x$

$$\frac{\text{New area}}{\text{Original area}} = \frac{(2x)^2}{x^2} = \frac{4x^2}{x^2} = 4 : 1$$

11. (b); Area of square = $\frac{1}{2} \times (\text{diagonal})^2$
 $= \frac{1}{2} \times 5.2 \times 5.2 = 13.52 \text{ cm}^2$

12. (b); Number of tin sheets = $\frac{100 \times 100}{20 \times 20} = 25$

13. (b); Number of Handkerchiefs = $\frac{175 \times 105}{35 \times 35} = 15$

14. (c); Let the side of square be 10.

Area of square = $10^2 = 100$

New side = $10 \times \frac{150}{100} = 15$

New area = $15 \times 15 = 225$

percent increase in area = $(225 - 100)\% = 125\%$

15. (b); Let the side of square be 'a'

diagonal of square = $\sqrt{2} a$

Required ratio = $\frac{a^2}{(\sqrt{2}a)^2} = \frac{a^2}{2a^2} = 1 : 2$

16. (b); Area of triangle = $9 \times y$

Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times a^2$

$$= \frac{\sqrt{3}}{4} \times 6 \times 6 = 9\sqrt{3}$$

According to question

$9 \times y = 9 \times \sqrt{3}, y = \sqrt{3}$

17. (b); Let the side of equilateral triangle be a

Given, $\frac{\sqrt{3}}{2} a = \sqrt{6}, a = 2\sqrt{2}$

Area = $\frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} \times 4 \times 2 = 2\sqrt{3}$

18. (b); Area = $\frac{1}{2} az + \frac{1}{2} ay + \frac{1}{2} tb + \frac{1}{2} cx$

$$= \frac{1}{2} [bt + cx + ay + az]$$

19. (a); Circumference = $2\pi r$

$$352 = 2\pi r, r = \frac{352}{2 \times 22} \times 7 = 56 \text{ m}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 56 \times 56 = 9856 \text{ m}^2$$

20. (a); Circumference of a circle = 88 cm

$$2\pi r = 88, r = \frac{88}{2 \times 22} \times 7, r = 14$$

Area of circle = $\pi r^2 = \pi(14)^2 = 196\pi$

21. (c); Area of regular hexagon = $6 \times \left(\frac{\sqrt{3}}{4} a^2 \right)$

$$= 6 \times \frac{\sqrt{3}}{4} \times 2 \times 2 = 6\sqrt{3} \text{ m}^2$$

22. (d); Required area = $[10 \times 2\pi - \pi(1)^2]$
 $= 20\pi - \pi = 19\pi \text{ m}^2$

23. (d); Required ratio = $\frac{\left(\frac{a}{2}\right)^2}{\left(\frac{\sqrt{2}a}{2}\right)^2} = 1 : 2$ a = side of square

24. (a); Area of circle = πr^2

Area of square = a^2

$$a^2 = \pi r^2, \frac{a^2}{r^2} = \frac{\pi}{1}, \frac{a}{r} = \sqrt{\pi} : 1$$

25. (d); Length of arc = 20, Circumference = $2\pi r$

Angle, $\theta = 60^\circ$

$$2\pi r \left(\frac{60^\circ}{360^\circ} \right) = 20, r = \frac{20 \times 6}{2\pi} = \frac{60}{\pi} \text{ cm}$$

26. (c); Let side of equilateral triangle be a

then its height = $\frac{\sqrt{3}}{2} a$

Required ratio = $\frac{a}{\frac{\sqrt{3}}{2} a} = 2 : \sqrt{3}$

27. (d); Area of rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} xy$

28. (a); Area of parallelogram = BC × height
 $= (50 \times 20) \text{ m}^2 = 1000 \text{ m}^2$

29. (c); Area of triangle = $\sqrt{S(S-a)(S-b)(S-c)}$
 Let $a = 15 \text{ m}$, $b = 16 \text{ m}$, $c = 17 \text{ m}$
 $S = \frac{a+b+c}{2} = 24$
 $= \sqrt{24(24-15)(24-16)(24-17)}$
 $= \sqrt{24 \times 9 \times 8 \times 7} = \sqrt{8 \times 3 \times 3 \times 3 \times 8 \times 7} = 24\sqrt{21}$

30. (c); Here, $S = \frac{20+21+29}{2} = 35 \text{ m}$
 Area = $\sqrt{35(35-20)(35-21)(35-29)}$
 $= \sqrt{35 \times 15 \times 14 \times 6} = 210 \text{ m}^2$

31. (a); Hypotenuse = 20 cm
 Perimeter = $2 \times 24 = 48 \text{ cm}$
 Let the length and breadth be l and b respectively
 $l + b + 20 = 48$, $l + b = 28$... (i)
 By pythagorus theorem,
 $l^2 + b^2 = 400$, $(l + b)^2 = 28^2$
 $l^2 + b^2 + 2lb = 784$, $2lb = 384$, $lb = 192$
 $l - b = \sqrt{784 - 4 \times 192} = \sqrt{16} = 4$... (ii)
 Solving (i) and (ii), we get
 $l = 16 \text{ cm}$ and $b = 12 \text{ cm}$

32. (c); Given hypotenuse = 8

$$\sqrt{2}a = 8, \quad a = \frac{8}{\sqrt{2}}$$

Area of isosceles triangle

$$= \frac{1}{2} \times \frac{8}{\sqrt{2}} \times \frac{8}{\sqrt{2}} = 16 \text{ cm}^2$$

33. (b); Area of parallelogram = $b \times h$
 $240 = b \times 12 \Rightarrow \text{base, } b = 20 \text{ cm}$

34. (c); Let x and y be the sides of the square

$$\text{then, } \frac{x^2}{y^2} = \frac{9}{1}, \quad \frac{x}{y} = \frac{3}{1}$$

$$\text{Ratio of perimeter} = \frac{4x}{4y} = \frac{x}{y} = \frac{3}{1} = 3:1$$

35. (c); Area of carpet = $\frac{105}{3.50} = 30 \text{ m}^2$

$$\text{width} = 5 \text{ m, length} = \frac{30}{5} = 6 \text{ m}$$

36. (d); Required = $\pi \left(\frac{d_1}{2} \right)^2 - \pi \left(\frac{d_2}{2} \right)^2$
 $= \pi(5)^2 - \pi(4)^2 = 25\pi - 16\pi = 9\pi \text{ cm}^2$

37. (b); Let the side be a and b of squares s_1 and s_2 respectively,

$$\text{Ratio of area} = \frac{a^2}{b^2} = \frac{4}{25}$$

$$\frac{6^2}{b^2} = \frac{4}{25}, \quad b^2 = \frac{6 \times 6 \times 25}{4}, \quad b = 15 \text{ cm}$$

38. (d); Let the width be x

$$\text{then length} = \frac{4}{3}x, \quad \text{Area} = \frac{4}{3}x \cdot x = 300$$

$$x^2 = \frac{900}{4}, \quad x = 15 \text{ cm}$$

$$\text{then length} = \frac{4}{3} \times 15 = 20 \text{ cm}$$

$$\text{Difference} = (20 - 15) \text{ cm} = 5 \text{ cm}$$

39. (d); Circumference = πd

$$\pi d - d = 105, \quad (\pi - 1)d = 105, \quad \frac{15}{7}d = 105$$

$$d = \frac{105}{15} \times 7 = 49 \text{ cm}$$

40. (d); Let r be the radius of circle and a be the side of square

$$\text{Side of square} = \sqrt{256} = 16 \text{ cm}$$

$$\text{Diameter of circle} = 16\sqrt{2}$$

$$\text{Radius of circle} = 8\sqrt{2}$$

$$\text{Area of circle} = \pi(8\sqrt{2})^2 = 128\pi \text{ cm}^2$$

Moderate

1. (c); Let the length and breadth be x and y respectively.

$$\text{Area} = xy = 600 \quad \dots(i)$$

$$\text{Perimeter} = 2(x + y) = 100$$

$$x + y = 50 \quad \dots(ii)$$

By using formula

$$(x - y)^2 = (x + y)^2 - 4xy, \quad (x - y)^2 = 50^2 - 4 \times 600$$

$$x - y = \sqrt{100} = 10, \quad x - y = 10 \quad \dots(iii)$$

On solving (ii) and (iii)

$$x = 30 \text{ and } y = 20$$

Length of plot = 30 m

2. (d); Area of park = $2w(l + b - 2w)$
 $= 2 \times 12(120 + 80 - 2 \times 12) = 24 \times 176 = 4224 \text{ m}^2$

3. (b); Perimeter of square = 80 cm

$$4a = 80, \quad a = 20 \text{ cm}$$

Perimeter of rectangle = 80 cm

$$2(l + b) = 80$$

$$l + b = 40 \text{ cm} \quad \dots(i)$$

According to question

$$a^2 - lb = 100, \quad 20^2 - lb = 100$$

$$lb = 400 - 100, \quad lb = 300$$

$$(l - b)^2 = 40^2 - 4 \times 300$$

$$l - b = \sqrt{400}$$

$$l - b = 20 \quad \dots(ii)$$

On solving (i) and (ii)

$$l = 30, \quad b = 10$$

length and breadth be 30 cm and 10 cm

4. (b); Let the shorter side be x

the longer side be y

$$x^2 + y^2 = 13^2 \quad \dots(i)$$

$$\frac{1}{2}xy = 30, \quad xy = 60 \quad \dots(ii)$$

On solving (i) and (ii)

$$x + y = \sqrt{289}, \quad x + y = 17$$

$$x - y = \sqrt{49}, \quad x - y = 7$$

Therefore, the length of shorter side is 5 cm.

5. (c); Let equal sides be ' x ' m each

$$\text{Then, } x^2 + x^2 = (50\sqrt{2})^2, \quad x = 50$$

$$\text{Perimeter of triangle} = 50 + 50 + 50\sqrt{2}$$

$$= 100 + 50\sqrt{2}$$

$$\text{Cost of fencing} = 3(100 + 50\sqrt{2}) = 300 + 150\sqrt{2}$$

$$= 300 + 212.13 = 512.13 > 500$$

6. (d); Let the side be $5x$, $5x$ and $4x$

$$\text{Perimeter} = 14$$

$$5x + 5x + 4x = 14$$

$$14x = 14, \quad x = 1$$

$$\text{Height of triangle} = \sqrt{5^2 - 2^2} = \sqrt{25 - 4} = \sqrt{21}$$

$$\text{Area} = \frac{1}{2} \times 4 \times \sqrt{21} = 2\sqrt{21} \text{ cm}^2$$

7. (d); Side of square $a = \frac{1}{\sqrt{2}} \times 12\sqrt{2} = 12 \text{ cm}$

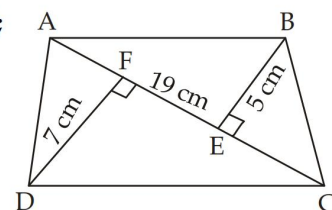
$$\text{Perimeter of square} = 4a = 4 \times 12 = 48 \text{ cm}$$

$$\text{Perimeter of equilateral triangle} = 48 \text{ cm}$$

$$\text{Side of equilateral triangle} = \frac{48}{3} = 16 \text{ cm}$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} (16)^2 = 64\sqrt{3} \text{ cm}^2$$

8. (b);



Area of quadrilateral ABCD

$$= \frac{1}{2} \times AC \times (BE + DF) = \frac{1}{2} \times 19 \times 12 = 114 \text{ cm}^2$$

9. (d); Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

$$\frac{\sqrt{3}}{4} a^2 = 5, \quad a^2 = \frac{20}{\sqrt{3}}, \quad a = \frac{20^{1/2}}{3^{1/4}}$$

$$\text{Perimeter of triangle} = 3 \times \frac{20^{1/2}}{3^{1/4}} = 3^{3/4} \cdot 20^{1/2}$$

$$\text{Now, } 2\pi R = 3^{3/4} \cdot 20^{1/2}$$

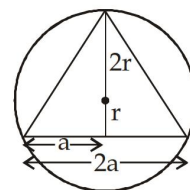
$$R = \left(\frac{3^{3/4} \cdot 20^{1/2}}{2\pi} \right) \text{ m}$$

10. (a); $(2a)^2 - a^2 = (3r)^2$

$$4a^2 - a^2 = 9r^2$$

$$3a^2 = 9r^2$$

$$\text{Radius of in circle, } r = \frac{1}{\sqrt{3}} a$$



Radius of circum circle, $2r = \frac{2}{\sqrt{3}}a$

$$A = \pi r^2 = \pi \left(\frac{1}{\sqrt{3}}a \right)^2 = \frac{\pi a^2}{3}$$

$$B = \pi (2r)^2 = \pi \left(\frac{2}{\sqrt{3}}a \right)^2 = \frac{4}{3}\pi a^2$$

11. (b); Side of square, $AB = 4$ cm

diagonal of square $= 4\sqrt{2}$ cm

$$\text{Radius of circle} = \frac{1}{2} \times 4\sqrt{2} = 2\sqrt{2} \text{ cm}$$

12. (c); Area of shaded region $= \frac{1}{2} \cdot \pi \cdot (14)^2 + 2 \times \frac{1}{2} \pi \cdot (7)^2$

$$= \frac{1}{2} \cdot \pi (14^2 + 2 \times (7)^2) \text{ cm}^2$$

$$= \frac{1}{2} \times \frac{22}{7} \times (196 + 98) = \frac{1}{2} \times \frac{22}{7} \times 294 = 462 \text{ cm}^2$$

13. (b); Required area $= 6 \times \text{area of equilateral triangle}$

$$= 6 \times \frac{\sqrt{3}}{4} a^2 = \frac{3\sqrt{3}}{2} a^2$$

14. (b); We know that the diagonals of a rhombus bisect each other at right angles.

In right angles $\triangle AOB$, we have

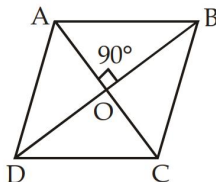
$$OA^2 = AB^2 - OB^2$$

$$= 5^2 - 4^2$$

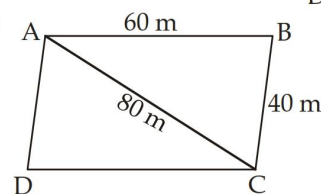
$$= 25 - 16 = 9$$

$$OA = 3$$

$$AC = 2 \times OA = 2 \times 3 = 6 \text{ m}$$



15. (c);



In $\parallel$ gm ABCD, $AB = 60$ m, $BC = 40$ m and $AC = 80$ m

$$S = \frac{1}{2}(60 + 40 + 80) = 90 \text{ m}$$

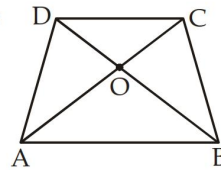
$$\text{Area of } \triangle ABC = \sqrt{S(S-a)(S-b)(S-c)}$$

$$= \sqrt{90 \times 30 \times 50 \times 10} = 300\sqrt{15} \text{ m}^2$$

$$\text{Area of } \parallel \text{ gm} = 2 \times \text{Area of } \triangle ABC = 2 \times 300\sqrt{15}$$

$$= 600\sqrt{15} \text{ m}^2$$

16. (d);



Given, $AB = 2CD$ and $AB \parallel CD$

In $\triangle AOB$ and $\triangle COD$

$\angle AOB = \angle COD$ (vertically opposite angles)

$\angle OAB = \angle OCD$ (alternate interior angles)

Therefore, $\triangle AOB$ and $\triangle COD$ are similar triangles.

$$\frac{\text{area}(\triangle AOB)}{\text{area}(\triangle COD)} = \frac{AB^2}{CD^2} = \frac{(2CD)^2}{CD^2} = 4:1$$

17. (a); Required area $= AB^2 + \frac{1}{2}BD \times CX$

$$= 2.5^2 + \frac{1}{2} \times 2.5 \times 2.16$$

$$= 6.25 + 2.5 \times 1.08 = 6.25 + 2.70 = 8.95 \text{ cm}^2$$

18. (b); Let the side of the triangle be $3x$, $4x$ and $5x$.

Perimeter $= 36$

$$3x + 4x + 5x = 36, \quad 12x = 36, \quad x = 3$$

then the sides are 9, 12 and 15

$$\text{Area of triangle} = \frac{1}{2} \times 9 \times 12 = 54 \text{ cm}^2$$

19. (c); Area of square $= (\text{side})^2 = x^2$

$$\text{Area of triangle} = \frac{1}{2} \times x \times h$$

According to question

$$\frac{1}{2} \times x \times h = x^2, \quad h = 2x$$

20. (c); Area of triangle $= 150 \text{ m}^2$

$$\frac{1}{2} \times 3x \times 4x = 150, \quad x^2 = \frac{150}{6} = 25, \quad x = 5$$

$$\text{Height of triangle} = 4x = 4 \times 5 = 20 \text{ m}$$

21. (b); Distance between longer sides $= 4$ m

Distance between shorter sides $= x$ m

$$\text{Area of parallelogram} = 8 \times 4 = 5 \times x$$

$$x = \frac{8 \times 4}{5} = \frac{32}{5} = 6.4 \text{ m}$$

22. (a); Let the base be ' x ' cm

its altitude $= 2x$ cm

$$\text{Area of parallelogram} = 72 \text{ cm}^2$$

$$x \cdot 2x = 72 \Rightarrow x^2 = 36, \quad x = 6 \text{ cm}$$

length of base $= 6$ cm

23. (b); Area of circle = πr^2

$$\pi r^2 = 154$$

$$r^2 = \frac{154}{22} \times 7 = 7^2, \quad r = 7$$

Perimeter of circles = Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

24. (b); Let the radius be 'r'

$$\pi r^2 = \pi(22^2 + 19^2 + 8^2), \quad r^2 = 484 + 361 + 64$$

$$r^2 = 909$$

Therefore radius is nearly equal to 30 cm.

25. (c); Area of rectangular carpet = 120 m^2

$$lb = 120$$

$$\text{Perimeter} = 46$$

$$2(l + b) = 46, \quad l + b = 23$$

$$(l + b)^2 = l^2 + b^2 + 2lb \Rightarrow l^2 + b^2 = 529 - 240$$

$$l^2 + b^2 = 289, \quad l^2 + b^2 = (17)^2 \Rightarrow l^2 + b^2 = d^2$$

The length of diagonal, $d = 17$

26. (d); Area = $lb = 60$

$$d^2 = l^2 + b^2 \Rightarrow d + l = 5b, \quad d = 5b - l$$

$$d^2 = 25b^2 + l^2 - 10bl$$

$$l^2 + b^2 = 25b^2 + l^2 - 10bl$$

$$24b^2 = 600, \quad b^2 = \frac{600}{24} = 25, \quad b = 5$$

$$l = \frac{60}{5} = 12 \text{ m}$$

27. (a); Let the diagonal of square be x .

$$\text{Then, area of square} = \frac{1}{2}x^2$$

$$\text{diagonal of new square} = 2x$$

$$\text{area of new square} = \frac{1}{2}(2x)^2 = \frac{1}{2} \times 4x^2 = 2x^2$$

Area of new square = $4 \times$ area of original square

Therefore, area of square becomes four-fold.

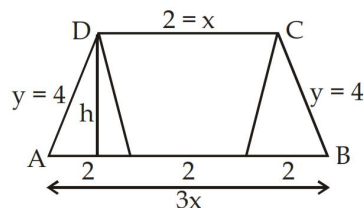
28. (c); Area of path = $2w(l + b + 2w)$

$$= 2 \times 1(24 + 14 + 2)$$

$$= 2 \times 40 = 80 \text{ m}^2$$

$$\text{Number of tiles} = \frac{80}{.2 \times .2} = 2000$$

29. (b);



Let ABCD be the isosceles trapezium

Given the sum of opposite sides are equal

$$DC + AB = BC + AD$$

According to question

$$4x = 2y \Rightarrow 2x = y$$

$$\text{and } 4x + 2y = 16 \Rightarrow 2x + y = 8$$

$$2y = 8 \Rightarrow y = 4 \text{ \& } x = 2$$

$$AB + CD = 8 \text{ cm}$$

$$\Rightarrow CD = 2 \text{ cm and } AD = 6 \text{ cm}$$

$$\text{Height of trapezium} = \sqrt{4^2 - 1^2} = \sqrt{15}$$

$$\text{Area of trapezium} = \frac{1}{2}(\sqrt{15})(2 + 6) = 4\sqrt{15} \text{ cm}^2$$

30. (c); Area of square = 22 cm^2

$$\text{Perimeter of square} = 4\sqrt{22} \text{ cm}$$

$$\text{Circumference of circle} = 2\pi r = 4\sqrt{22}$$

$$r = \frac{2\sqrt{22}}{\pi}$$

$$\text{Area of circle} = \pi r^2 = \pi \left(\frac{2\sqrt{22}}{\pi} \right)^2 = \frac{\pi \times 4 \times 22}{\pi^2}$$

$$= \frac{4 \times 22}{\pi} = \frac{4 \times 22 \times 7}{22} = 28 \text{ cm}^2$$

31. (b); Area of path = $w(l + b - w) = 3(80 + 60 - 3)$

$$= 3 \times 137 = 411 \text{ m}^2$$

32. (a); Circumference of wheel = $2 \times \frac{22}{7} \times 1.75 \text{ m} = 11 \text{ m}$

$$\text{No. of revolutions} = \frac{11000}{11} = 1000$$

33. (d); Diagonal of rectangle = 25 cm

$$\text{length} = 20 \text{ cm}$$

$$l^2 + b^2 = d^2, \quad 400 + b^2 = 625$$

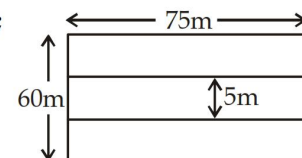
$$b^2 = 225, \quad b = 15 \text{ cm}$$

$$\text{Area of rectangle} = l \times b = 20 \times 15 = 300 \text{ cm}^2$$

34. (a); Percentage increase in area = $10 + 5 + \frac{10 \times 5}{100}$

$$= 15 + 0.5 = 15.5\%$$

35. (c);



$$\text{Area of the way} = \text{width of way} \times \text{length of garden}$$

$$= 5 \times 75 = 375 \text{ m}^2$$

36. (c); Let the length and width of rectangle be x and y respectively.

Then, area = xy

$$\text{New length} = x \times \frac{160}{100} = \frac{8x}{5}$$

When area remains same

$$\frac{8}{5}x \times \text{new width} = xy$$

$$\text{New width} = \frac{5}{8}y$$

$$\text{decrease in width} = y - \frac{5}{8}y = \frac{3}{8}y$$

$$\begin{aligned} \text{Percentage decrease in width} &= \frac{3y \times 100}{8 \times y} = \frac{75}{2}\% \\ &= 37\frac{1}{2}\% \end{aligned}$$

37. (a); Let the radius of circle be r .

$$\text{area of circle} = \pi r^2$$

When radius is decreased by ' n ' the area becomes half

$$\text{New radius} = r - n, \quad \text{New area} = \pi(r - n)^2$$

$$\pi(r - n)^2 = \frac{1}{2}\pi r^2, \quad (r - n)^2 = \frac{1}{2}r^2, \quad r - n = \frac{1}{\sqrt{2}}r$$

$$r - \frac{1}{\sqrt{2}}r = n, \quad \left(\sqrt{2} - 1\right)r = \sqrt{2}n, \quad r = \frac{\sqrt{2}n}{\sqrt{2} - 1}$$

38. (c); Circumference = $2\pi r = 2 \times \frac{22}{7} \times 56 = 352$ cm

Perimeter of square = 352

$$4a = 352, \quad a = \frac{352}{4} = 88$$

39. (a); Area of path = $w(l + b - w)$
 $= 10(150 + 120 - 10) = 10 \times 260 = 2600$ m²
 Cost of gravelling = $2 \times 2600 = \text{Rs. } 5200$

40. (d); Perimeter of garden = $4a = 400$
 $4a = 400, \quad a = 100$
 Area of garden = $a^2 = (100)^2 = 10^4$

Difficult

1. (a); Area of garden = $\pi(12)^2 = 144\pi$ m²
 Area of circular pool = $\pi(5)^2 = 25\pi$ m²

According to question

$$\pi(12 + r)^2 - \pi(12)^2 = 25\pi$$

$$(12 + r)^2 - 144 = 25, \quad (12 + r)^2 = 169$$

$$(12 + r)^2 = 13^2, \quad 12 + r = 13, \quad r = 1 \text{ m}$$

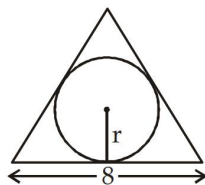
2. (c); Area of square = $(\text{side})^2 = (21)^2 = 441$ cm²

Area of circle = πr^2 , Here, $r = 10.5$

$$\text{Area} = \pi \times (10.5)^2 = \frac{22}{7} \times 10.5 \times 10.5 = 346.50 \text{ cm}^2$$

$$\text{Area of shaded portion} = 441 - 346.50 = 94.5 \text{ cm}^2$$

3. (b);



Side of equilateral triangle = 8 cm

$$\text{radius of in circle, } r = \frac{1 \times 8}{2\sqrt{3}} = \frac{4}{\sqrt{3}} \text{ cm}$$

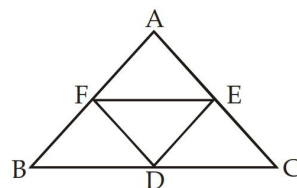
$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4}(\text{side})^2$$

$$= \frac{\sqrt{3}}{4} \times 8 \times 8 = 16\sqrt{3} \text{ cm}^2$$

$$\text{Area of circle} = \pi r^2 = \frac{22}{7} \times \left(\frac{4}{\sqrt{3}}\right)^2 = 16.76 \text{ cm}^2$$

$$\begin{aligned} \text{Area of remaining portion} &= (16\sqrt{3} - 16.76) \\ &= 10.95 \text{ cm}^2 \end{aligned}$$

4. (c); Let ABC be the triangle and the triangle formed by joining the mid point of the triangle is DEF.



Therefore the area of $\triangle DEF$ is equal to $\frac{1}{4}$ of the area of $\triangle ABC$.

$$\text{area of } \triangle ABC = \sqrt{S(S-a)(S-b)(S-c)}$$

$$S = \frac{a+b+c}{2} = \frac{12+18+26}{2} = 28 \text{ cm}$$

$$\begin{aligned}\text{Area} &= \sqrt{28 \times 16 \times 10 \times 2} \\ &= \sqrt{7 \times 2 \times 2 \times 4 \times 4 \times 5 \times 2 \times 2}\end{aligned}$$

$$\text{Area of } \triangle ABC = 16\sqrt{35} \text{ cm}^2$$

$$\begin{aligned}\text{Area of } \triangle DEF &= \frac{1}{4} \times \text{ar}(\triangle ABC) \\ &= \frac{1}{4} \times 16\sqrt{35} = 4\sqrt{35} \text{ cm}^2\end{aligned}$$

5. (d); Area of path = $w(l + b - w)$
 $= 2(19.25 + 12.50 - 2)$
 $= 2(31.75 - 2) = 2 \times 29.75 = 59.50 \text{ m}^2$
 Cost of paving = Rs. 12.32×59.50 = Rs. 733.04

6. (a); Perimeter of shaded region
- $$\begin{aligned}&= \frac{1}{2} \times 2\pi(\text{OQ}) + \frac{1}{2} \times 2\pi\left(\frac{1}{2}\text{OQ}\right) + \frac{1}{2} \times 2\pi\left(\frac{1}{2}\text{OQ}\right) \\ &= \pi \times 14 + \frac{\pi}{2} \times 14 + \frac{\pi}{2} \times 14 \\ &= \pi(14 + 7 + 7) = \frac{22}{7} \times 28 = 88 \text{ cm}\end{aligned}$$

7. (b); Circumference of a circular field = 440 m

$$2\pi r_1 = 440, \quad r_1 = \frac{440 \times 7}{2 \times 22} = 70 \text{ m}$$

$$r_2 = (70 + 10) \text{ m} = 80 \text{ m}$$

$$\begin{aligned}\text{Area of circular path} &= \pi(r_2^2 - r_1^2) \\ &= \pi(r_2 - r_1)(r_2 + r_1) = \pi(80 - 70)(80 + 70) \\ &= \pi \times 10 \times 150 = 1500\pi \text{ m}^2\end{aligned}$$

$$\text{cost of cultivating the path} = \frac{70}{100} \times 1500\pi$$

$$= \frac{70}{100} \times 1500 \times \frac{22}{7} = \text{Rs. } 3300$$

8. (a); Area of square park = $(\text{side})^2 = (100)^2 = 10000 \text{ m}^2$

$$\text{Area of 4 flower bed} = 4 \times \frac{1}{4} \pi r^2 = \pi r^2$$

$$= \frac{22}{7} \times 14 \times 14 = 616 \text{ m}^2$$

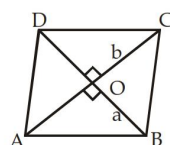
$$\begin{aligned}\text{Area of remaining part of park} &= (10000 - 616) \text{ m}^2 \\ &= 9384 \text{ m}^2\end{aligned}$$

9. (c); Perimeter of a rhombus is $2P$

$$\text{Let } OB = b, \quad OC = a$$

$$\text{side of Rhombus} = \frac{2P}{4} = \frac{P}{2}$$

In right $\triangle OBC$



$$a^2 + b^2 = \frac{P^2}{2} \Rightarrow 4a^2 + 4b^2 = P^2 \quad \dots (i)$$

Also,

$$2a + 2b = m \quad \text{on squaring}$$

$$4a^2 + 4b^2 + 8ab = m^2$$

$$4a^2 + 4b^2 = m^2 - 8ab \quad \dots (ii)$$

$$m^2 - 8ab = P^2 \Rightarrow 4 \times 2ab = m^2 - P^2$$

$$2ab = \frac{1}{4}(m^2 - P^2)$$

$$\Rightarrow \text{area of Rhombus} = \frac{1}{4}(m^2 - P^2)$$

10. (a); Area of rectangular park = $60 \times 40 = 2400 \text{ m}^2$

$$\text{Area of lawn} = 2109 \text{ m}^2$$

$$\text{Area of the road} = 2400 - 2109 = 291 \text{ m}^2$$

$$\text{Area of road} = w(l + b - w)$$

$$291 = w(60 + 40 - w), \quad 291 = w(100 - w)$$

$$w^2 - 100w + 291 = 0$$

$$w^2 - 97w - 3w + 291 = 0$$

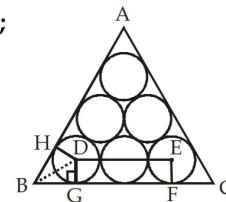
$$w(w - 97) - 3(w - 97) = 0$$

$$(w - 3)(w - 97) = 0$$

$$w \neq 97 \text{ is not possible}$$

Therefore the width of the road is 3 m.

11. (c);



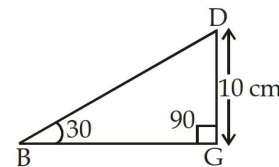
Here, the radius of each circle is 10 cm

and DEFG is a rectangle

$$DE = 10 + 20 + 10 = 40 \text{ cm}$$

$$DE = GF = 40 \text{ cm}$$

To find BG we take $\triangle BGD$



Every angle of equilateral triangle is 60° .

Therefore $\angle DBG = 30^\circ$

$$\tan 30^\circ = \frac{DG}{BG}, \quad BG = \frac{DG}{\tan 30^\circ} = \frac{10}{\frac{1}{\sqrt{3}}} = 10\sqrt{3}$$

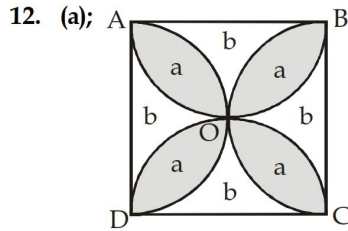
$$\text{and } BG = FC = 10\sqrt{3}$$

$$BC = BG + GF + FC = 10\sqrt{3} + 40 + 10\sqrt{3}$$

$$= 40 + 20\sqrt{3} = 20(2 + \sqrt{3})$$

$$\text{Perimeter of triangle} = 3 \times 20(2 + \sqrt{3})$$

$$= 60(2 + \sqrt{3})$$



$$\text{Area of square} = (\text{side})^2 = 8^2 = 64$$

$$\text{Area of shaded region} = 4a$$

$$\text{Area of remaining region} = 4b$$

$$\text{Therefore, } 4a + 4b = 64$$

$$a + b = 16 \quad \dots(i)$$

$$\text{Area of semicircle } AOB = \frac{1}{2}\pi(4)^2$$

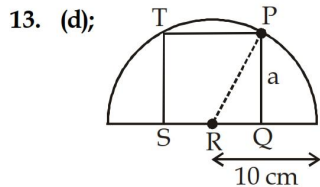
$$2a + b = 8\pi \quad \dots(ii)$$

On solving (i) and (ii)

$$a = 8\pi - 16 = 8(\pi - 2)$$

$$\text{Area of shaded region, } 4a = 4 \times 8(\pi - 2)$$

$$= 32(\pi - 2) \text{ cm}^2$$



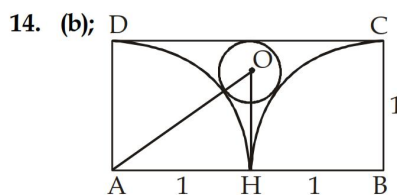
Let the side of square be 'a'

$$QR = \frac{a}{2}, PQ = a$$

$$PR^2 = PQ^2 + QR^2$$

$$10^2 = a^2 + \frac{a^2}{4}, 100 = \frac{5a^2}{4}, a^2 = 80, a = 4\sqrt{5} \text{ cm}$$

$$\text{Perimeter of square} = 4a = 4 \times 4\sqrt{5} = 16\sqrt{5} \text{ cm}$$



Let the radius of smaller circle be r.

then in $\triangle AOH$,

$$AO^2 = OH^2 + AH^2, (1 + r)^2 = (1 - r)^2 + 1^2,$$

$$1 + r^2 + 2r = 1 + r^2 - 2r + 1, r = \frac{1}{4}$$

Therefore, area of shaded region

$$= 2 \times 1 - \frac{1}{2}\pi(1)^2 - \pi\left(\frac{1}{4}\right)^2$$

$$= 2 - \frac{22}{14} - \frac{22}{112} = 2 - \frac{11}{7} - \frac{11}{56}$$

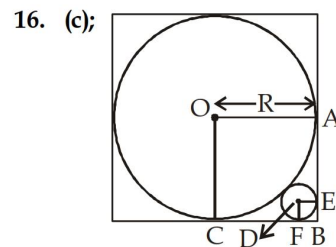
$$= \frac{112 - 88 - 11}{56} = \frac{13}{56} \text{ square units.}$$

15. (b); Let $AB = BC = a$, then $AC = \sqrt{2}a$

$$AC^2 = 128, AC = \sqrt{128}, AC = 8\sqrt{2}, a = 8 \text{ cm}$$

$$\text{Area of semicircles} = \frac{1}{2}\pi\left(\frac{a}{2}\right)^2 + \frac{1}{2}\pi\left(\frac{a}{2}\right)^2$$

$$= \frac{1}{2}\pi\left(\frac{a^2}{4} + \frac{a^2}{4}\right) = \frac{1}{2}\pi\frac{a^2}{2} = \frac{64}{4}\pi = 16\pi \text{ cm}^2$$



Here, $OA = OC = R$

and $\angle O = \angle A = \angle C = \angle B = 90^\circ$

Therefore OABC is a square

$$OA = AB = R$$

$$OB^2 = OA^2 + AB^2, OB = \sqrt{2} R$$

Similarly, DFBE is also a square

$$\text{and, } DB = \sqrt{2} r$$

$$OB = OD + DB$$

$$\sqrt{2} R = (R + r) + \sqrt{2} r, (\sqrt{2} - 1)R = (\sqrt{2} + 1)r$$

$$r = \left(\frac{\sqrt{2} - 1}{\sqrt{2} + 1}\right)R, r = (3 - 2\sqrt{2})R$$

17. (d); Area of quadrant AOBCE = $\frac{1}{4}\pi r^2$

$$= \frac{1}{4} \times \frac{22}{7} \times 3.5 \times 3.5 = 9.625 \text{ cm}^2$$

$$\text{Area of } \triangle AOD = \frac{1}{2} \times OA \times OD$$

$$= \frac{1}{2} \times 3.5 \times 2 = 3.5 \text{ cm}^2$$

$$\text{Area of shaded portion} = (9.625 - 3.5) \text{ cm}^2 = 6.125 \text{ cm}^2$$

18. (d); Diameter = PS = $2 \times r = 2 \times 6$, PS = 12

$$PQ = QR = RS = \frac{12}{3} = 4 \text{ cm}$$

$$\begin{aligned} \text{Area of shaded region} &= \frac{\pi}{2} 6^2 - \frac{\pi}{2} 4^2 \\ &= 18\pi - 8\pi = 10\pi \end{aligned}$$

$$\begin{aligned} \text{Area of unshaded region} &= \frac{\pi}{2} 6^2 + \frac{\pi}{2} 4^2 \\ &= 18\pi + 8\pi = 26\pi \end{aligned}$$

$$\frac{\text{Area of shaded region}}{\text{Area of unshaded region}} = \frac{10\pi}{26\pi} = \frac{5}{13} = 5:13$$

19. (b); Sum of internal angle of a hexagon = $(n - 2)180$
= 720°

Where n = no. of sides

Which means two circle of same radius

Therefore the area of shaded region = $2\pi r^2$

20. (a); Let AD = 6 units then

$$AB = BC = CD = \frac{6}{3} = 2 \text{ units}$$

Area of shaded region

$$= \frac{1}{2} \pi [(2)^2 - (1)^2] + \frac{1}{2} \times \pi [2^2 - 1^2]$$

$$= \frac{1}{2} \times \pi \times 3 + \frac{1}{2} \times \pi \times 3 = 3\pi$$

$$\text{Area of whole circle} = \pi(3)^2 = 9\pi$$

$$\frac{\text{Area of shaded region}}{\text{Area of whole circle}} = \frac{3\pi}{9\pi} = 1:3$$

Previous Year (Memory Based)

1. (c); Let each side of lawn be 'a'.

$$\text{Area of path} = 2w(2a + 2w)$$

$$196 = 2 \times 4 (2a + 8)$$

$$2a + 8 = 24.5$$

$$2a = 16.5, a = 8.25 \text{ m}$$

2. (d); Let the diagonal of other square be x.

$$\text{Area of first square} = \frac{1}{2} \times 4\sqrt{2} \times 4\sqrt{2} = 16 \text{ cm}^2$$

$$\text{Area of other square} = 2 \times 16 = 32 \text{ cm}^2$$

$$\frac{1}{2} x^2 = 32$$

$$x^2 = 64, x = 8 \text{ cm.}$$

3. (c); Let the diagonal of two square having area A and B be $2x$ and $5x$ respectively.

Ratio of area be

$$\frac{A}{B} = \frac{\frac{1}{2}(2x)^2}{\frac{1}{2}(5x)^2} = \frac{4}{25} = 4:25$$

4. (b); Area of field diminished by = $\left(x + y - \frac{xy}{100}\right)\%$

$$= \left(40 + 40 - \frac{40 \times 40}{100}\right)\% = 64\%$$

Area of field diminished by 64%

5. (b); Area of path = $2w(l + b - 2w)$

$$600 = 2w(70 - 2w)$$

$$4w^2 - 140w + 600 = 0$$

$$w^2 - 35w + 150 = 0$$

$$(w - 30)(w - 5) = 0$$

$$w = 5, 30 \text{ since } w \neq 30$$

$$\text{width of park} = 5 \text{ m.}$$

6. (a); Let the breadth be 'x'.

$$\text{length} = 5x$$

$$\text{Total area of plot} = 2 \times 245$$

$$5x^2 = 2 \times 245 \Rightarrow x^2 = 2 \times 49 \Rightarrow x = 7\sqrt{2}$$

$$\text{length of plot} = 5 \times 7\sqrt{2} = 35\sqrt{2}$$

7. (c); Let the length of sides of triangle are $4x$, $5x$ and $6x$.

H $\rightarrow$ Height corresponding to largest side

Area of triangle = rs

where r = radius of in circle

s = semi perimeter of triangle

$$s = \frac{4x + 5x + 6x}{2} \Rightarrow 7.5x$$

$$\text{Area} = 3 \times 7.5x$$

Area corresponding to largest side as base

$$22.5x = 1/2 \times 6x \times H \Rightarrow H = 7.5 \text{ unit}$$

8. (a); Possible values of perimeter

$$= 15 + 15 + 22 \text{ or } 15 + 22 + 22 = 52 \text{ or } 59$$

9. (c); When area remain same then base is decrease by 'x'.

$$0 = \left(10 - x - \frac{10x}{100}\right)\%$$

$$\frac{110}{100}x = 10, x = \frac{1000}{110} = 9\frac{1}{11}\%$$

10. (c); D is the mid point of AB and E is the mid point of AC.

$$\text{Therefore } DE = \frac{1}{2}BC$$

$$\frac{\Delta ADE}{\Delta ABC} = \frac{DE^2}{BC^2} = \frac{1}{4}$$

$$\begin{aligned} \text{Area of trapezium BCED} &= \Delta ABC - \Delta ADE \\ &= 4 - 1 = 3 \text{ units} \end{aligned}$$

$$\text{Required percentage} = \frac{3}{4} \times 100 = 75\%$$

11. (d); Let the base and altitude be $3x$ and $4x$

$$\text{respectively Area of triangle} = \frac{1}{2} \times 3x \times 4x$$

$$1176 = 6x^2 \Rightarrow x^2 = 196 \Rightarrow x = 14 \text{ cm}$$

$$\text{Altitude of a triangle} = 4x = 4 \times 14 = 56 \text{ cm.}$$

12. (a); Let the length of parallel side be $5x$ and $3x$.

$$\begin{aligned} \text{Area of trapezium} &= \frac{1}{2} (\text{sum of parallel sides} \times \\ &\text{distance between them}) \end{aligned}$$

$$1440 = \frac{1}{2} (5x + 3x) \times 24$$

$$1440 = \frac{1}{2} \times 8x \times 24$$

$$x = \frac{1440}{4 \times 24} = 15$$

$$\text{Length of longer side} = 15 \times 5 = 75 \text{ cm}$$

13. (a); Let the original radius be 'r'

$$\text{Area of circle} = \pi r^2$$

$$\pi(r+1)^2 = \pi r^2 + 22 \Rightarrow \pi(r+1)^2 - \pi r^2 = 22$$

$$1 + 2r = \frac{22}{22} \times 7$$

$$(2r+1) = 7 \Rightarrow 2r = 6, r = 3 \text{ cm}$$

14. (b); Perimeter of semicircle = $(\pi r + 2r)$

$$= \pi \times \frac{63}{2} + 2 \times \frac{63}{2} = \frac{63}{2} \times \frac{22}{7} + 63$$

$$= 99 + 63 = 162 \text{ cm}$$

15. (a); Circumference of circular park = 176

$$2\pi r = 176$$

$$r = \frac{176 \times 7}{2 \times 22} = 28$$

$$\text{Radius of park with road} = 28 + 7 = 35 \text{ m}$$

$$\text{Area of road} = \pi(35)^2 - \pi(28)^2$$

$$\frac{22}{7}(7)(63) = 1386 \text{ m}^2$$

16. (b); Let the radius circle be 'r'.

According to question

$$2\pi r - 2r = 30$$

$$2r\left(\frac{22}{7} - 1\right) = 30 \Rightarrow 2r = \frac{30}{15} \times 7 \Rightarrow r = 7 \text{ cm}$$

17. (c); Let the inner radius be 'r'

Then outer radius, $R = r + 5$

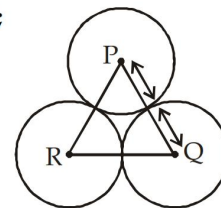
$$\frac{\text{Circumference of outer circular path}}{\text{Circumference of inner circular path}} = \frac{23}{22}$$

$$\frac{2\pi R}{2\pi r} = \frac{23}{22} \Rightarrow \frac{R}{r} = \frac{23}{22}$$

$$\frac{r+5}{r} = \frac{23}{22}, r = 110$$

$$d = 2r, d = 2 \times 110, d = 220 \text{ m.}$$

18. (b);



Triangle formed by joining the centres of the coin will be equilateral triangle.

$$\text{Area of } \Delta PQR = \frac{\sqrt{3}}{2} (\text{side})^2$$

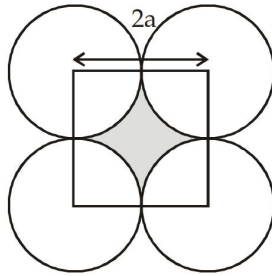
$$= \frac{\sqrt{3}}{4} \times 2^2 = \sqrt{3} \text{ cm}^2$$

Now, area of three sectors formed by PQR.

$$= 3 \times \frac{60}{360} \times \pi \times 1^2 = \pi/2$$

$$\text{Area of remaining portion} = \left(\sqrt{3} - \frac{\pi}{2} \right) \text{cm}^2$$

19. (b);



Area of shaded part = Area of square - Area of circle

$$= (2a)^2 - \pi a^2 = 4a^2 - \frac{22}{7}a^2 = \left(\frac{28 - 22}{7} \right) a^2 = \frac{6}{7}a^2$$

20. (b); Circumference of inner circle = 88 cm

$$\text{radius, } r = \frac{88}{2 \times 22} \times 7 = 14 \text{ cm}$$

Circumference of outer circle = 132 cm

$$\text{Radius, } R = \frac{132}{2 \times 22} \times 7 = 21 \text{ cm}$$

$$\text{area of ring} = \pi(R^2 - r^2)$$

$$= \pi(R - r)(R + r) = \frac{22}{7} \times 7 \times 35 = 770 \text{ cm}^2$$

21. (a); Radius of Incircle = $\frac{24}{2\sqrt{3}} = \frac{12}{\sqrt{3}}$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2$$

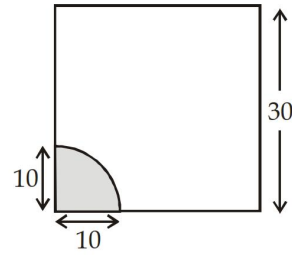
$$= \frac{1.732}{4} \times (24)^2 = 249.41$$

$$\text{Area of circle} = \pi r^2 = \pi \left(\frac{12}{\sqrt{3}} \right)^2$$

$$= \frac{22}{7} \times \frac{12 \times 12}{3} = 150.86$$

$$\text{Required area} = 249.41 - 150.86 = 98.55 \text{ cm}^2$$

22. (a);



$$\text{Horse can garze in} = \frac{90^\circ}{360^\circ} \times \pi r^2$$

$$= \frac{1}{4} \times \frac{22}{7} \times 10 \times 10 = 78.5 \text{ m}^2$$

23. (c); Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

$$\frac{\sqrt{3}}{4} a^2 = 4\sqrt{3}, a^2 = 4^2, a = 4 \text{ cm}$$

$$\text{Height of equilateral triangle} = \frac{\sqrt{3}}{2} a$$

$$h = \frac{\sqrt{3}}{2} \times 4, h = 2\sqrt{3} \text{ cm}$$

$$\text{Side of square} = 2\sqrt{3} \text{ cm}$$

$$\text{diagonal of square} = \sqrt{2} \times 2\sqrt{3} = 2\sqrt{6} \text{ cm.}$$

24. (a); Let the side of square be 'a'.

$$\text{Diagonal of square} = \sqrt{2} a.$$

$$\text{Area of square} = a^2$$

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} (\sqrt{2} a)^2$$

$$= \frac{\sqrt{3}}{4} \times 2 \times a^2 = \frac{\sqrt{3}}{2} a^2$$

$$\text{Required ratio} = \frac{\frac{\sqrt{3}}{2} a^2}{a^2} = \frac{\sqrt{3}}{2} = \sqrt{3} : 2.$$

25. (b); Let the radius of circle be 'r'.

side of square be 'a'

Area of square = Area of circle

$$a^2 = \pi r^2$$

$$\frac{a^2}{r^2} = \frac{\pi}{1} \Rightarrow \frac{a}{r} = \frac{\sqrt{\pi}}{1} = \sqrt{\pi} : 1$$

26. (d); Let the side of square be 'a'.

$$\text{Perimeter} = 4a = 160 \text{ m}^2, a = 40 \text{ m}.$$

$$\text{area of square} = 40^2 = 1600$$

Let the length and breadth be l and b respectively

$$2(l + b) = 160$$

$$l + b = 80 \quad \dots (i)$$

$$(\text{Area of square}) - (\text{Area of rectangle}) = 100$$

$$1600 - l \times b = 100$$

$$lb = 1500 \quad \dots (ii)$$

On solving (i) and (ii)

$$l - b = 20 \quad \dots (iii)$$

from (i) and (iii)

$$2l = 100, l = 50 \text{ m}$$

27. (d); Let the side of square be 'a' cm

$$\text{length of rectangle} = (a + 5) \text{ cm}$$

$$\text{breadth of rectangle} = (a - 3) \text{ cm}$$

According to question

$$\text{Area of square} = \text{Area of rectangle}$$

$$a^2 = (a + 5)(a - 3)$$

$$a^2 = a^2 - 3a + 5a - 15$$

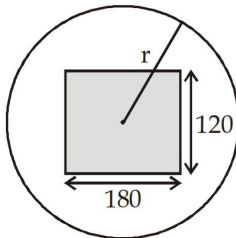
$$2a = 15, a = \frac{15}{2}$$

$$\text{Perimeter of rectangle} = 2(l + b)$$

$$= 2(a + 5 + a - 3) = 2(2a + 2) = 4\left(\frac{15}{2} + 1\right)$$

$$= 2 \times 17 = 34 \text{ cm}.$$

28. (c);



$$\text{Area of tank} = 180 \times 120 = 21600 \text{ m}^2$$

$$\text{Total area of circular plot} = 40000 + 21600 = 61600 \text{ m}^2$$

$$\text{Area of field} = \pi r^2 = 61600$$

$$r^2 = \frac{61600 \times 7}{22}$$

$$r = \sqrt{2800 \times 7} = 2 \times 70 = 140 \text{ m}$$

29. (b); Perimeter of rectangular table = 28 m

$$2(l + b) = 28$$

$$l + b = 14 \quad \dots (i)$$

$$\text{Area} = lb = 48 \quad \dots (ii)$$

On solving (i) and (ii)

$$(l - b)^2 = 196 - 192$$

$$l - b = 2 \quad \dots (iii)$$

$$2l = 16, l = 8$$

$$2b = 12, b = 6$$

$$\text{length of diagonal} = \sqrt{l^2 + b^2} = \sqrt{8^2 + 6^2}$$

$$= \sqrt{64 + 36} = \sqrt{100} = 10 \text{ m}$$

30. (d); Let side of square be 'a' cm

$$\text{Area} = a^2 = 484 \Rightarrow a = 22 \text{ cm}.$$

$$\text{Perimeter of square} = 4a = 4 \times 22 = 88 \text{ cm}$$

$$\text{circumference} = 88, 2\pi r = 88$$

$$r = \frac{88 \times 7}{2 \times 22} = 14 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

31. (c); Let the sides of 5 square be a, b, c, d and e respectively

$$4a = 24, a = 6 \text{ cm}$$

$$4b = 32, b = 8 \text{ cm}$$

$$4c = 40, c = 10 \text{ cm}$$

$$4d = 76, d = 19 \text{ cm}$$

$$4e = 80, e = 20 \text{ cm}$$

$$\text{Sum of areas of 5 square} = a^2 + b^2 + c^2 + d^2 + e^2$$

$$= 6^2 + 8^2 + 10^2 + 19^2 + 20^2$$

$$= 36 + 64 + 100 + 361 + 400$$

$$\text{Area of bigger square (side)}^2 = 961 \text{ cm}^2.$$

$$\text{side} = 31 \text{ cm}$$

$$\text{Perimeter} = 4 \text{ side} = 4 \times 31 = 124 \text{ cm}$$

32. (b); Area of path = $2w(l + b - 2w)$

$$600 = 2w(38 + 32 - 2w)$$

$$300 = w(70 - 2w)$$

$$150 = w(35 - w)$$

$$w^2 - 35w + 150 = 0$$

On solving above equation, we get

$$w = 5$$

33. (a); Let the side of the square be 'a'

$$\text{breadth of rectangle} = \frac{3}{2}a$$

$$\text{Area of rectangle} = 3 \times \text{Area of square}$$

$$20 \times \frac{3}{2}a = 3a^2 \Rightarrow 30a = 3a^2 \Rightarrow a = 10 \text{ cm}$$

34. (c); Let the breadth be x

$$\text{length} = x + 3, \text{ area} = 70$$

$$x(x + 3) = 70 \Rightarrow x^2 + 3x - 70 = 0$$

$$x^2 + 10x - 7x - 70 = 0, (x + 10) - 7(x + 10) = 0$$

$$x = -10, 7$$

Therefore the breadth be 7m

and length = $(7 + 3) = 10$ m

$$\text{Perimeter of floor} = 2(10 + 7) = 34 \text{ m}$$

35. (c); Circumference = $2\pi r$

$$\text{Distance covered in 1min} = 2 \times \frac{8}{40} \times \frac{22}{7} \times r$$

$$\text{New circumference} = 2 \times \pi \times r \times 10$$

$$\text{Time taken} = \frac{2\pi r \times 10 \times 40}{2\pi r \times 8} = 50 \text{ min.}$$

36. (c); Perimeter of triangle = 30 cm.

$$\text{Area} = 30 \text{ cm}^2$$

From triplet

$$(5, 12, 13) \text{ or } (3, 4, 5)$$

Where largest side is 13.

$$5^2 + 12^2 = 13^2$$

$$\text{and perimeter} = 5 + 12 + 13 = 30 \text{ cm}$$

$$\text{Area} = \frac{1}{2} \times 5 \times 12 = 30 \text{ cm}^2$$

Since both conditions are satisfied

Smallest side = 5 cm

37. (b); Distance covered by wheel = 2km 26 decameters

$$= 2260 \text{ m}$$

Distance covered in the revolution

$$= \frac{2260}{113} = 20 \text{ m}$$

$$\text{Circumference} = 20 \text{ cm}$$

$$\pi d = 20$$

$$d = \frac{20 \times 7}{22} = \frac{70}{11} = 6 \frac{4}{11} \text{ m}$$

38. (c); Old circumference = 4π

$$2\pi r = 4\pi, r = \frac{4\pi}{2\pi} = 2 \text{ cm}$$

$$\text{old area} = \pi(2)^2 = 4\pi \text{ cm}^2$$

$$\text{New circumference} = 8\pi$$

$$2\pi r = 8\pi$$

$$R = \frac{8\pi}{2\pi} = 4 \text{ cm}$$

$$\text{New area} = 16\pi \text{ cm}^2$$

So, the area becomes 4 times of the old area.

39. (d); According to question

Circumference of circle = area of circle

$$2\pi r = \pi r^2 \Rightarrow r = 2$$

$$\text{Diameter} = 2r = 2 \times 2 = 4$$

40. (c); Let the breadth be x m.

$$\text{Then length} = (x + 20) \text{ m}$$

According to question

$$x(x + 20) = (x - 10)(x + 50)$$

$$x^2 + 20x = x^2 + 50x - 10x - 500$$

$$x = 25 \Rightarrow \text{Breadth}$$

$$\text{Length} = 35$$

$$\text{Area of floor} = 35 \times 25 = 875 \text{ m}^2$$